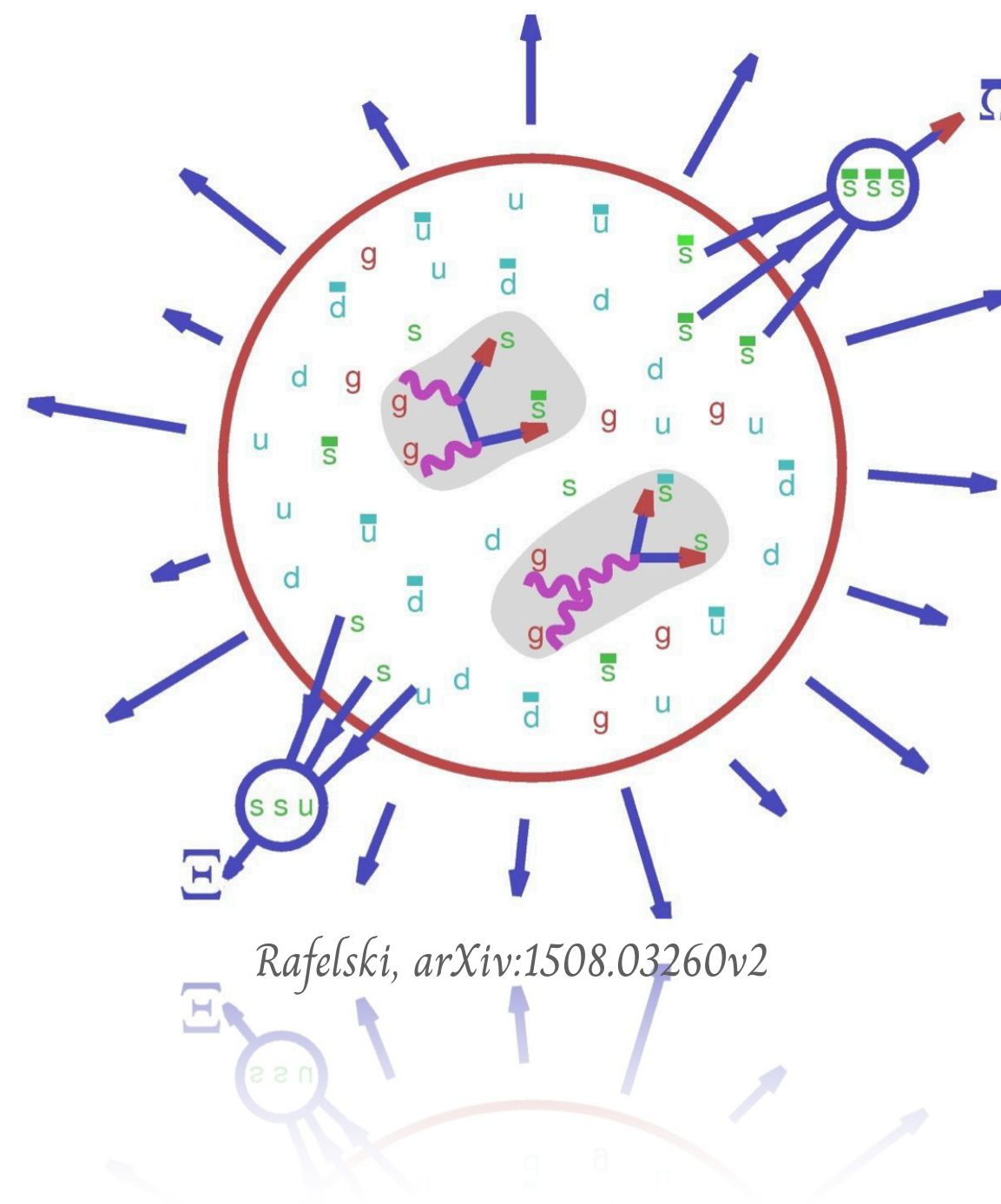
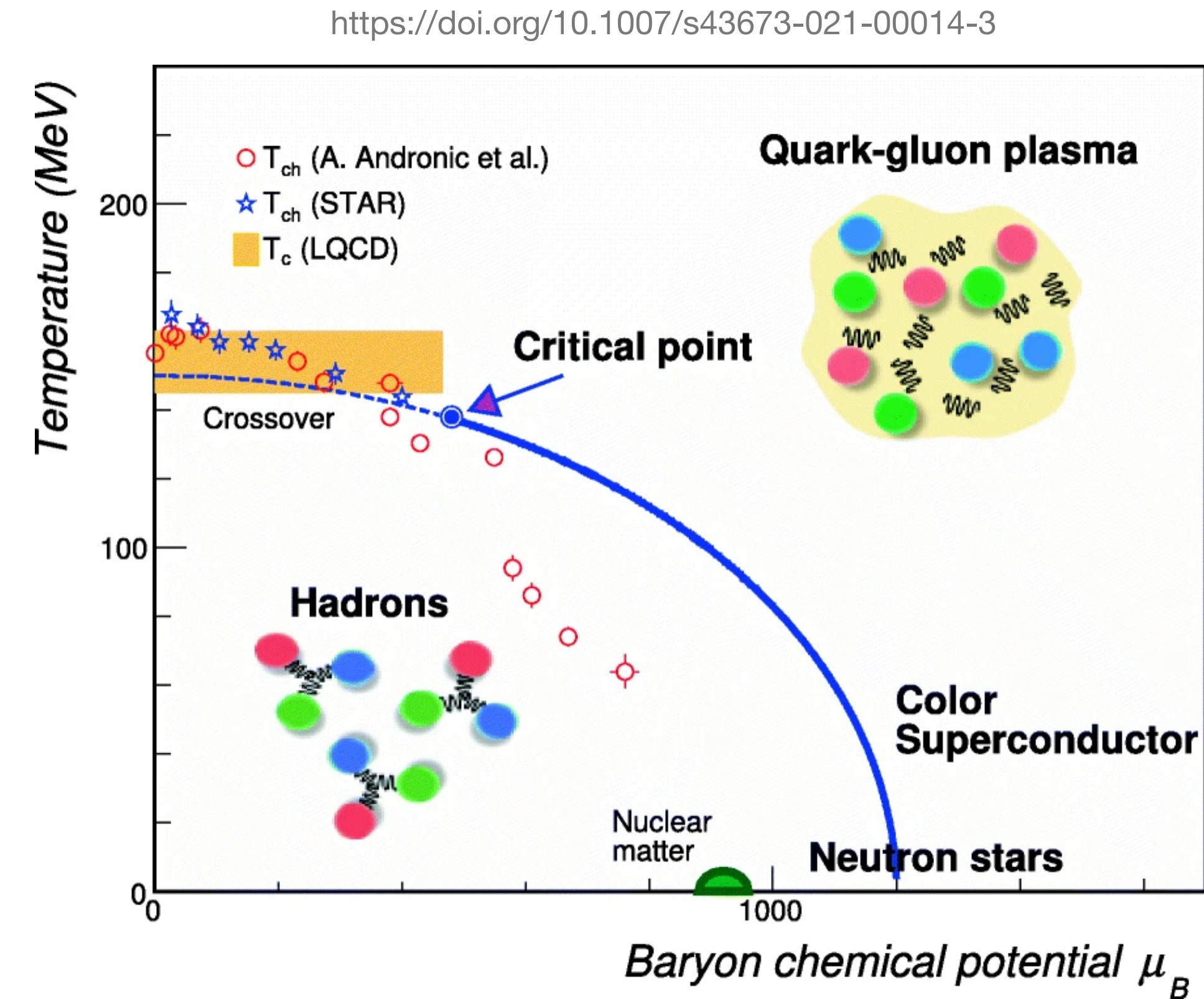
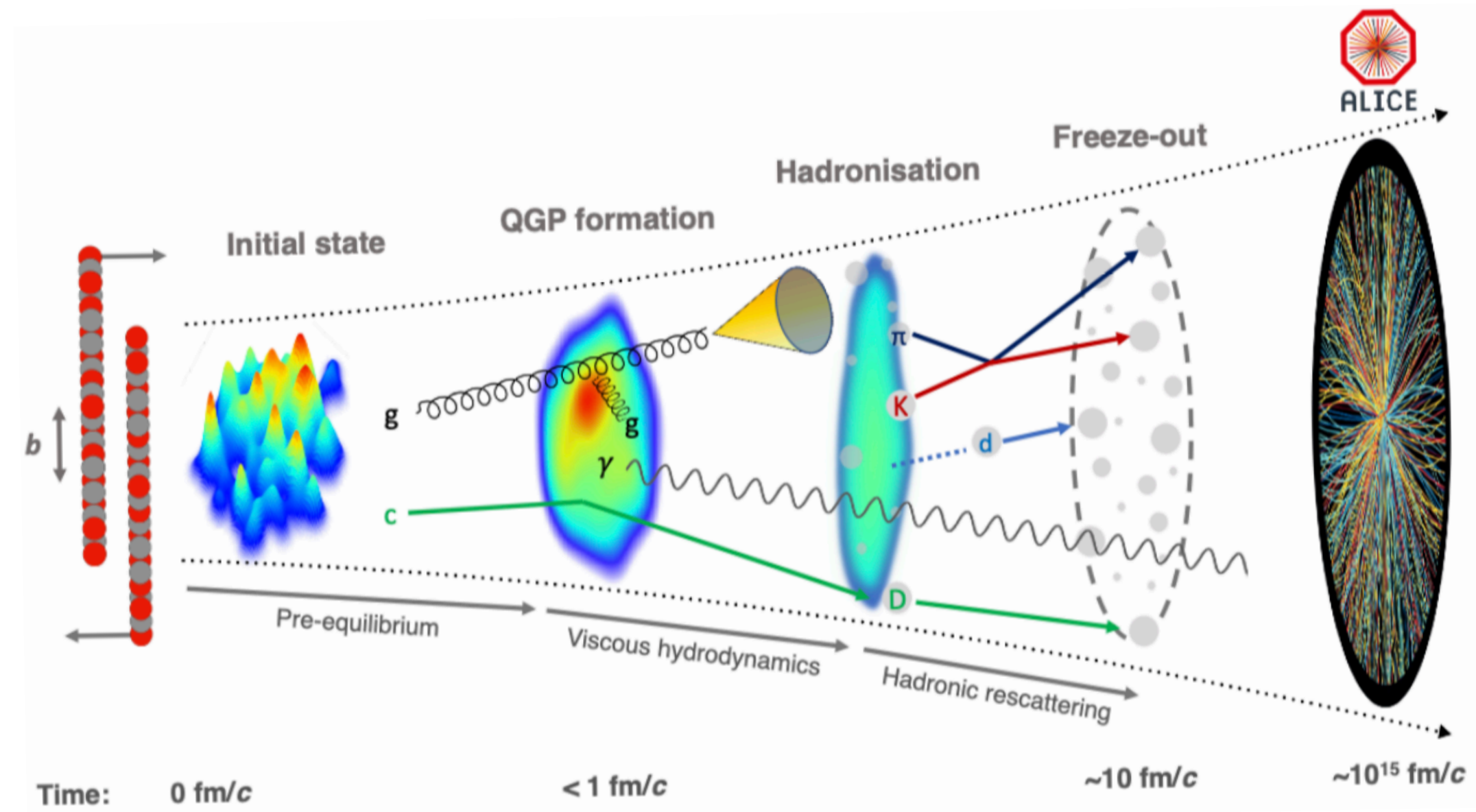


Strangeness Enhancement Across Collision Systems

D. A. Abdallah
Wayne State University



QCD Matter at Extreme Conditions

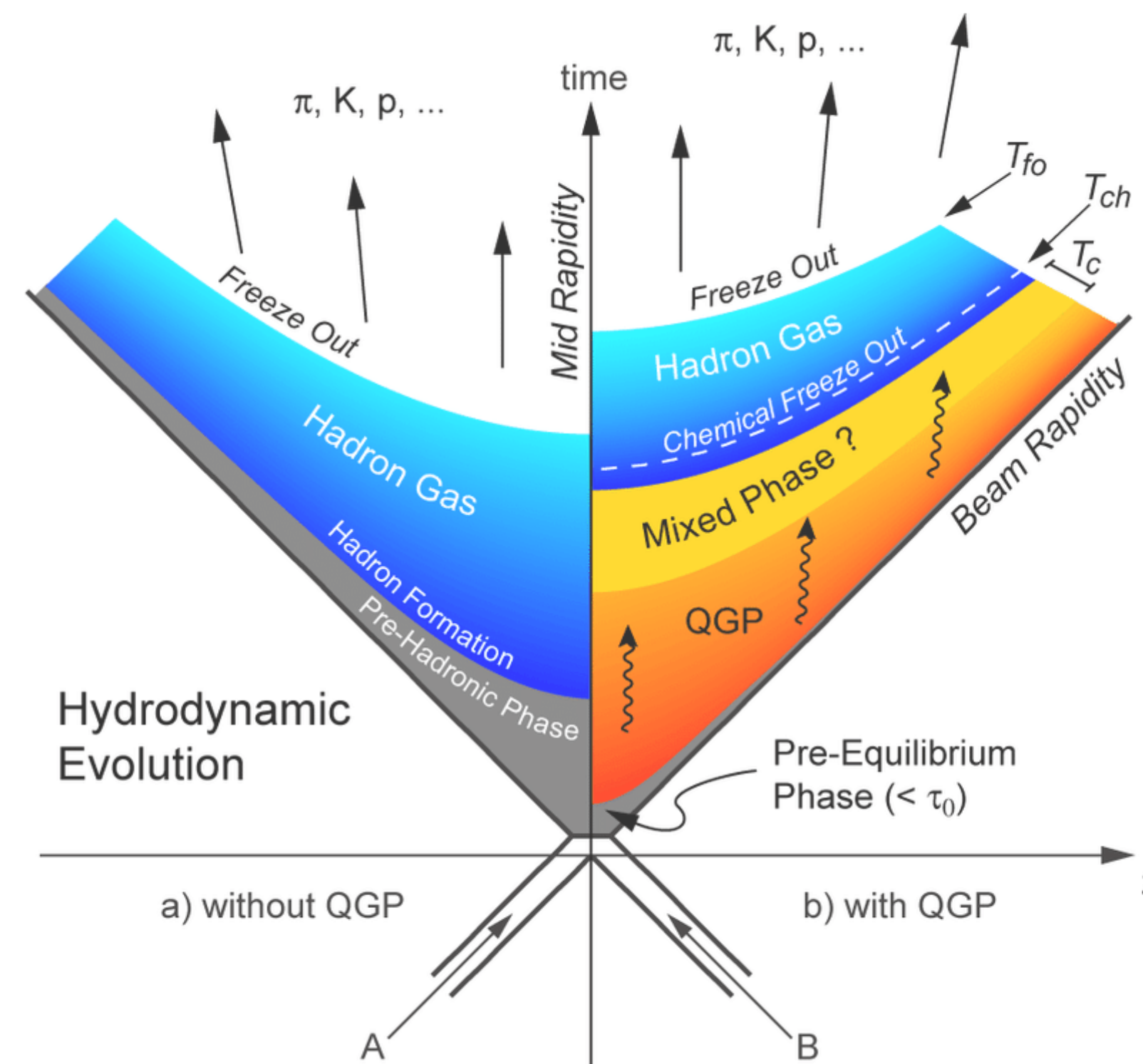


Signatures of the Quark-Gluon Plasma

Indirect probes of the short-lived QGP:

- ❖ Jet quenching
- ❖ Collective flow
- ❖ Quarkonia suppression
- ❖ Electromagnetic probes
- ❖ Strangeness enhancement **Today's topic**

Why Strangeness Enhancement is a QGP Signature ?

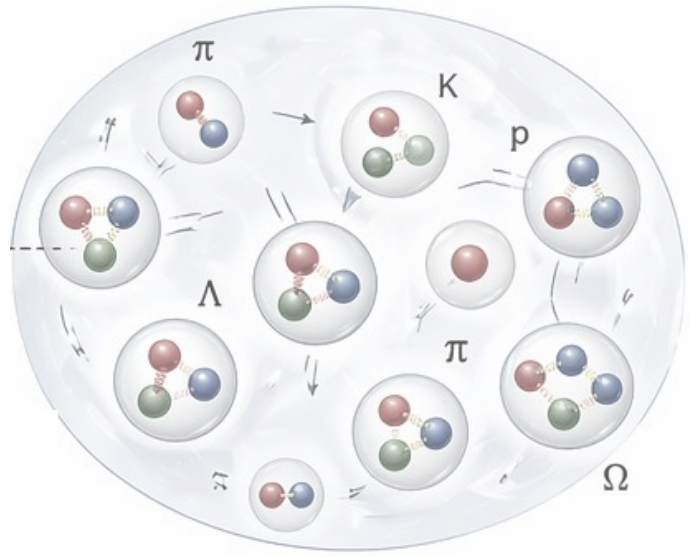


[10.1016/j.nuclphysa.2019.02.006](https://doi.org/10.1016/j.nuclphysa.2019.02.006)

Why Strangeness Enhancement is a QGP Signature ?

Hadronic medium

Hadrons (bound states of quarks)

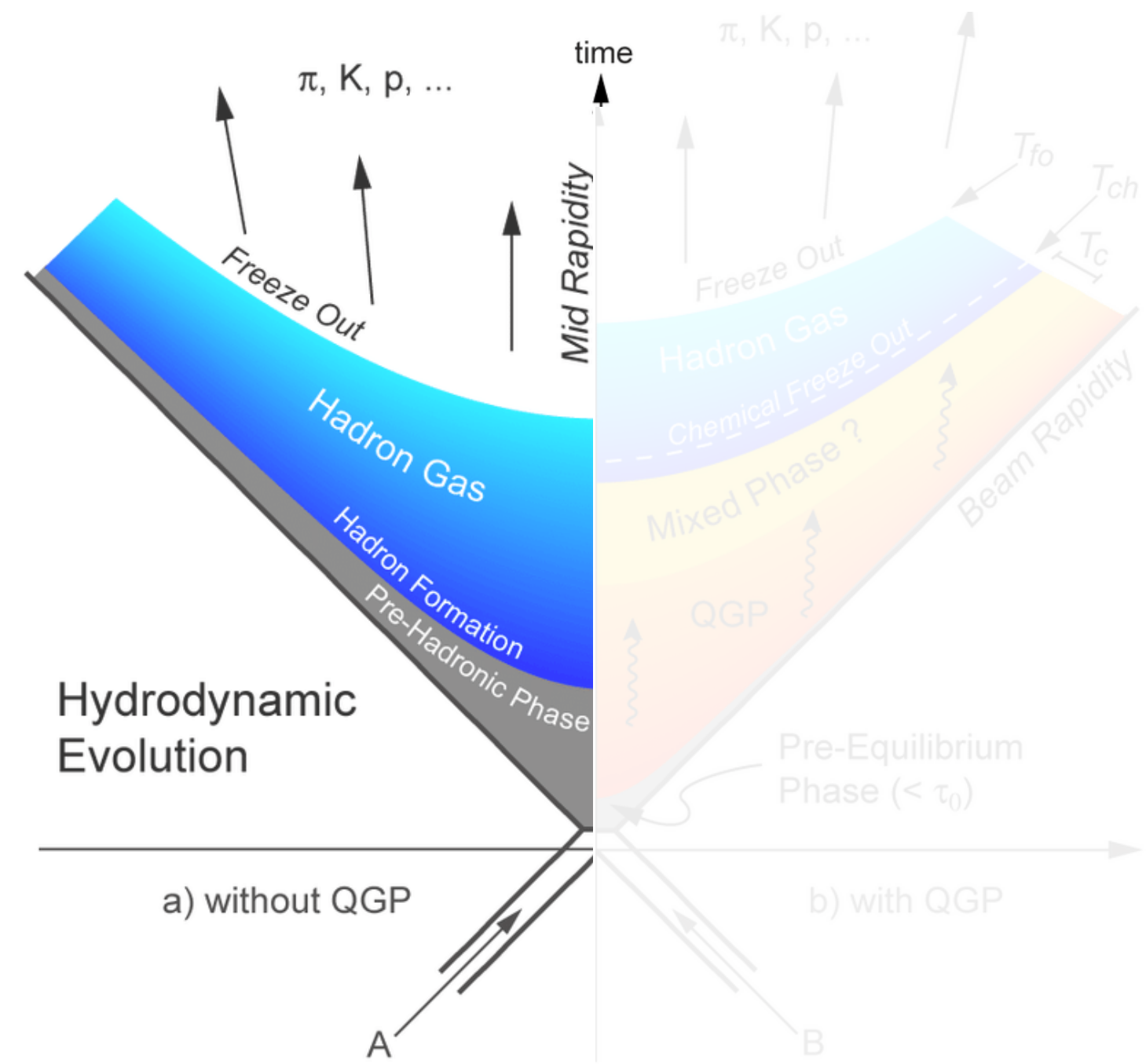


❖ Multi-step interactions



❖ Inefficient (threshold suppressed)

⇒ Suppressed strange quark production

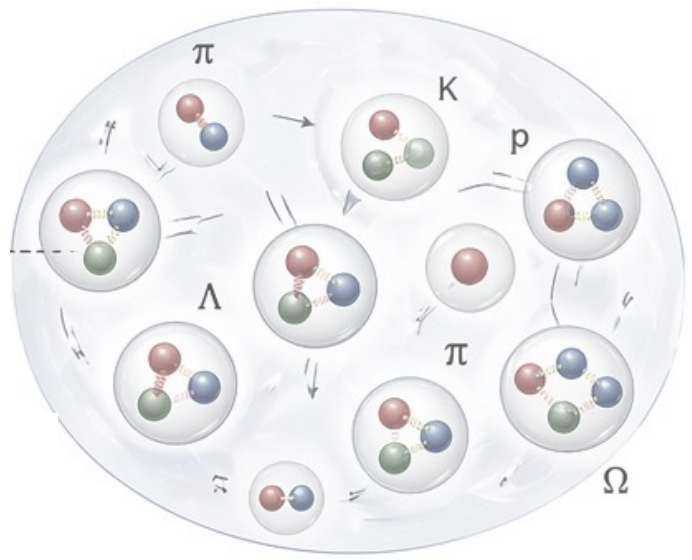


[10.1016/j.nuclphysa.2019.02.006](https://doi.org/10.1016/j.nuclphysa.2019.02.006)

Why Strangeness Enhancement is a QGP Signature ?

Hadronic medium

Hadrons (bound states of quarks)



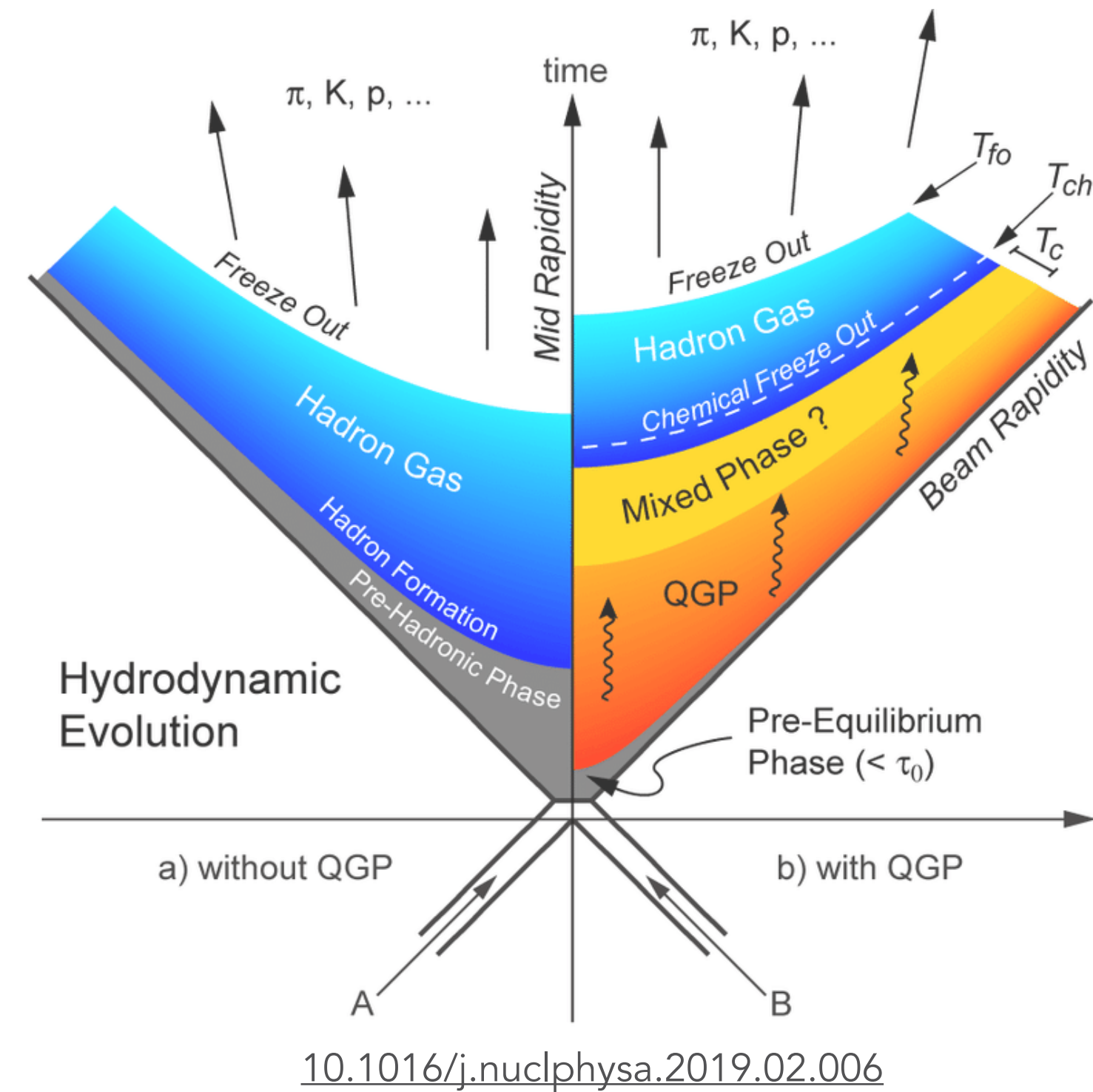
❖ Multi-step interactions

$$\pi\pi \rightarrow KK^- \quad \pi N \rightarrow \Lambda K$$

❖ Inefficient (threshold suppressed)

⇒ Suppressed strange quark production

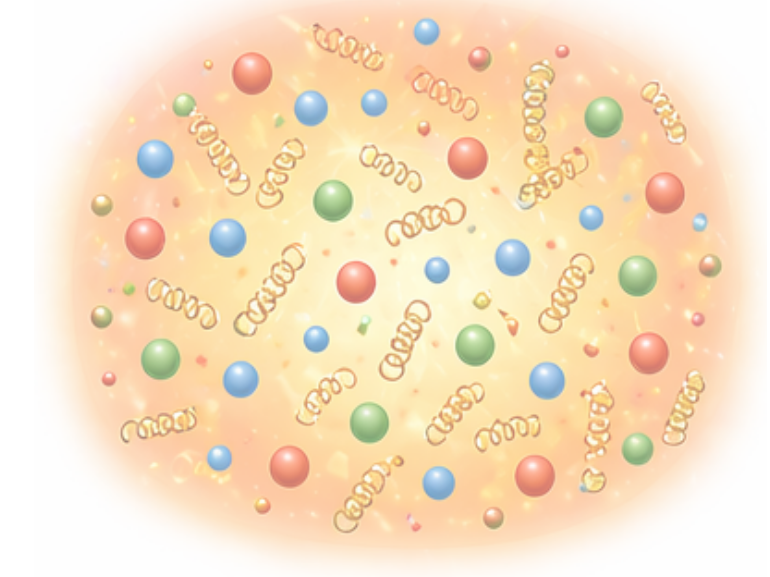
Vs.



10.1016/j.nuclphysa.2019.02.006

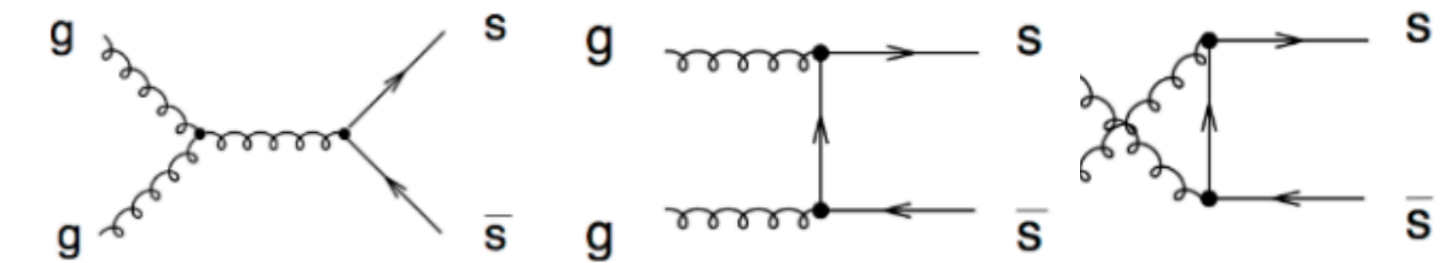
Partonic (QGP) medium

Deconfined quarks and gluons

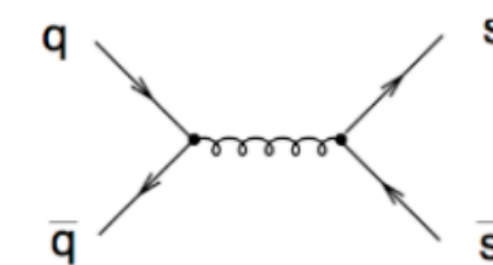


❖ Deconfined quarks + abundant gluons

$$gg \rightarrow s\bar{s}$$



$$q\bar{q} \rightarrow s\bar{s}$$

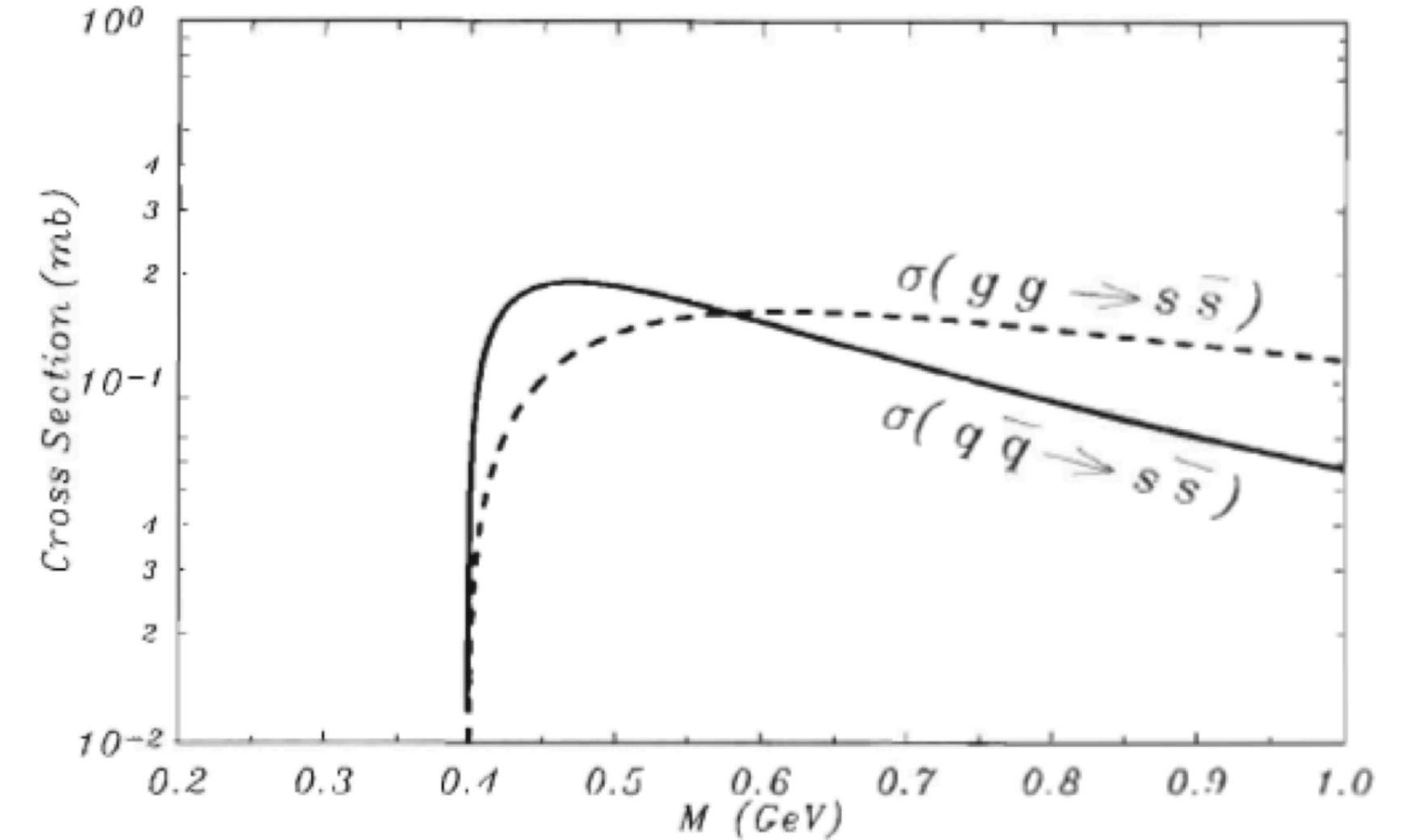


∴ Efficient production in QGP

⇒ Enhanced strange hadron yields

Strangeness Enhancement as a QGP Signature

- ❖ Rafelski & Müller (1982) proposed strangeness enhancement as evidence for QGP formation.
- ❖ Found that strangeness production in QGP is dominated by $gg \rightarrow s\bar{s}$.



Rafelski, J. 1983, <https://cds.cern.ch/record/147788/files/198311105.pdf>

VOLUME 48, NUMBER 16

PHYSICAL REVIEW LETTERS

19 APRIL 1982

Strangeness Production in the Quark-Gluon Plasma

Johann Rafelski and Berndt Müller

Institut für Theoretische Physik, Johann Wolfgang Goethe-Universität, D-6000 Frankfurt am Main, Germany

(Received 11 January 1982)

Rates are calculated for the processes $gg \rightarrow s\bar{s}$ and $u\bar{u}, d\bar{d} \rightarrow s\bar{s}$ in highly excited quark-gluon plasma. For temperature $T \geq 160$ MeV the strangeness abundance saturates during the lifetime ($\sim 10^{-23}$ sec) of the plasma created in high-energy nuclear collisions. The chemical equilibration time for gluons and light quarks is found to be less than 10^{-24} sec.

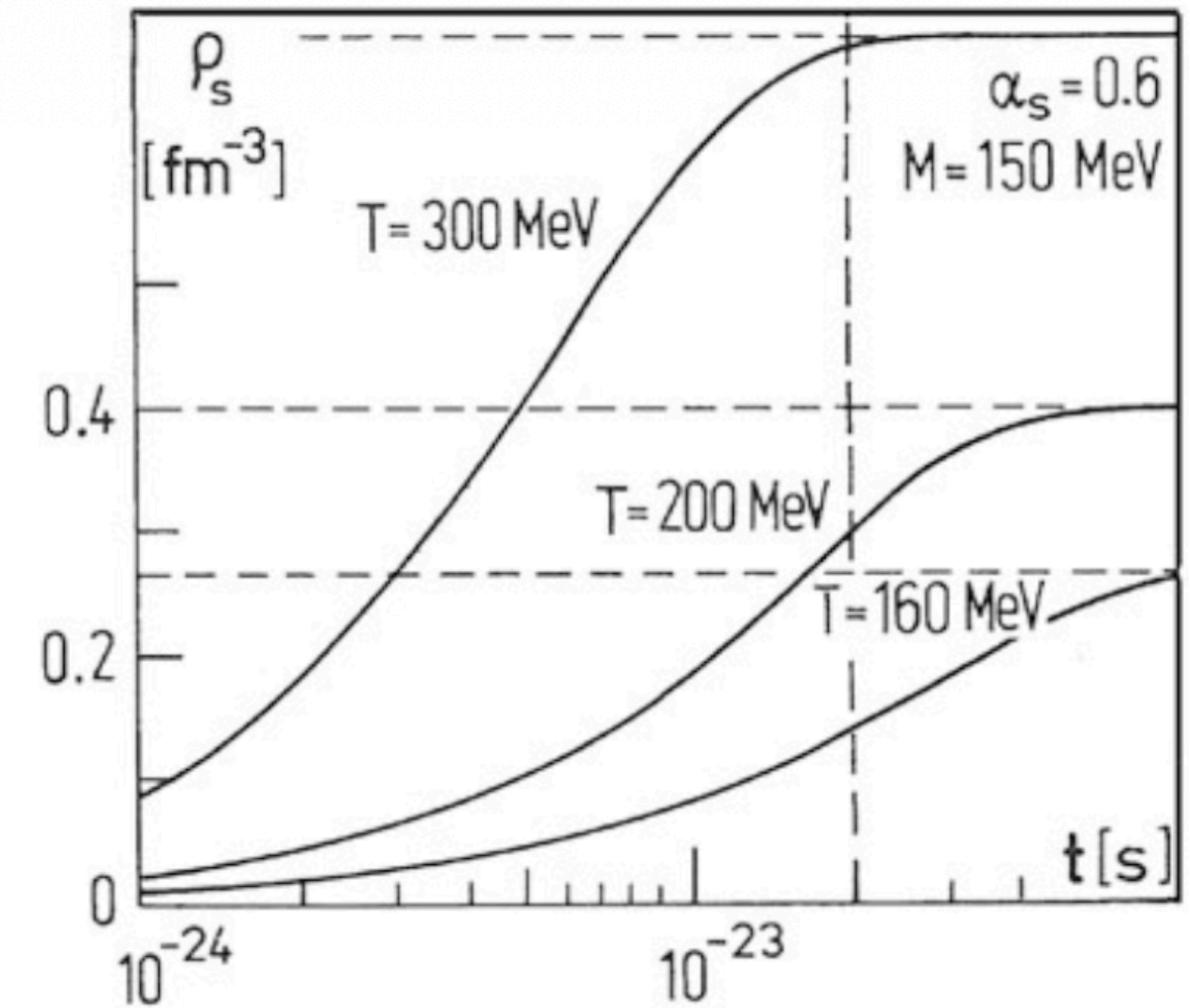
PACS numbers: 12.35.Ht, 21.65.+f

Strangeness Equilibration in QGP

❖ Strangeness density rises and saturates during plasma lifetime. $\tau_s < \tau_{QGP}$

⇒ Leads to enhanced strange hadron yields after hadronization.

❖ Final yields preserve memory of QGP phase.



10.1007/978-3-319-17545-4_31

Experimental Evidence in Heavy-Ions

- ❖ First observed at CERN SPS and later confirmed at RHIC and LHC.

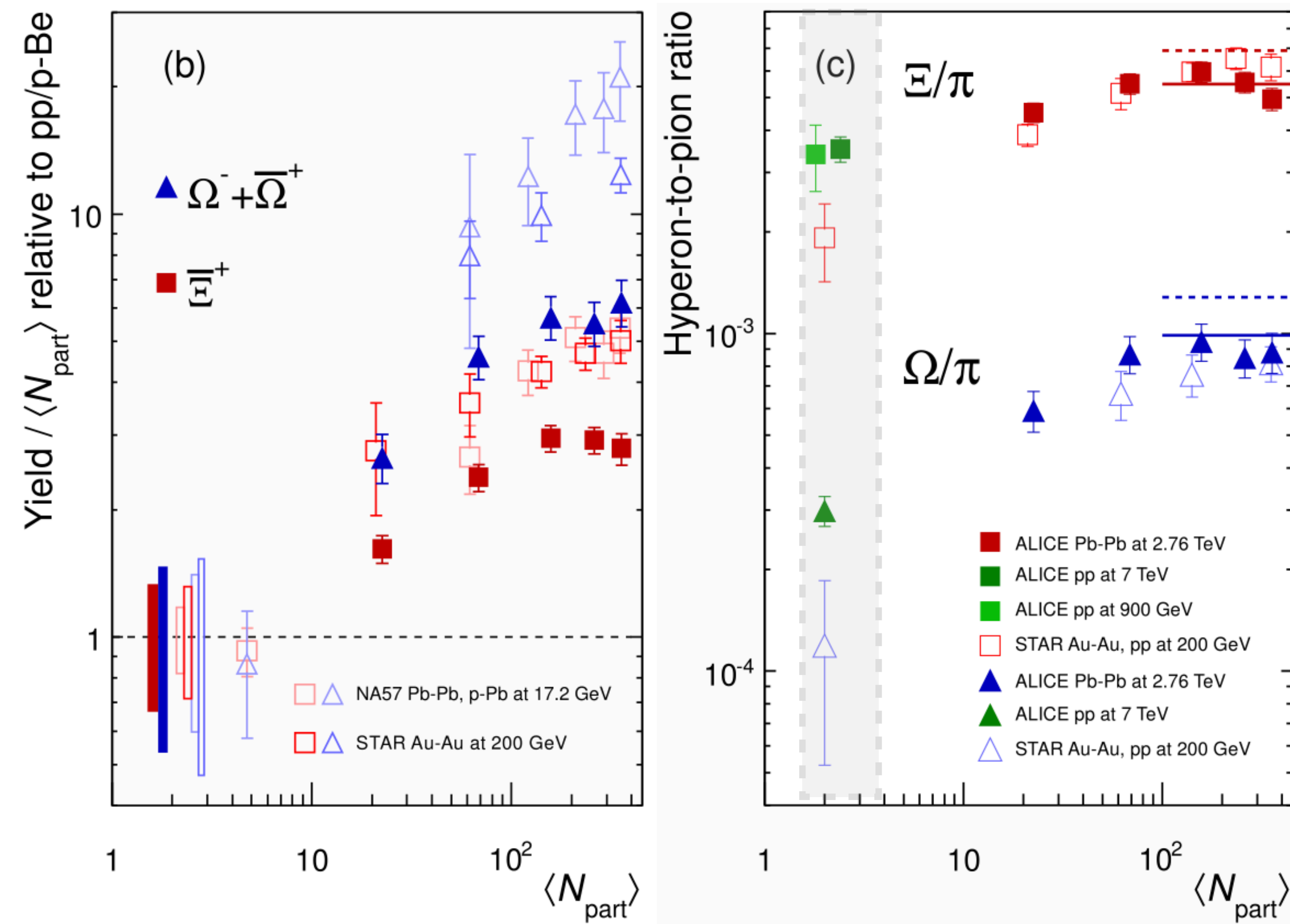
NA49 Coll., NA57 Coll., J. Phys. G 37 (2010)
 WA97 Coll., Phys. Lett. B 449 (1999)
 STAR Coll., Phys. Rev. C 77 (2008)

- ❖ Enhancement increases with system size and strange-quark content.

$$|S_{\Xi^\pm}| = 2 < |S_{\Omega^\pm}| = 3$$

$$\Xi^- = |dss\rangle \quad \Omega^- = |sss\rangle$$

- ❖ Approximate scaling across collision energies and systems!

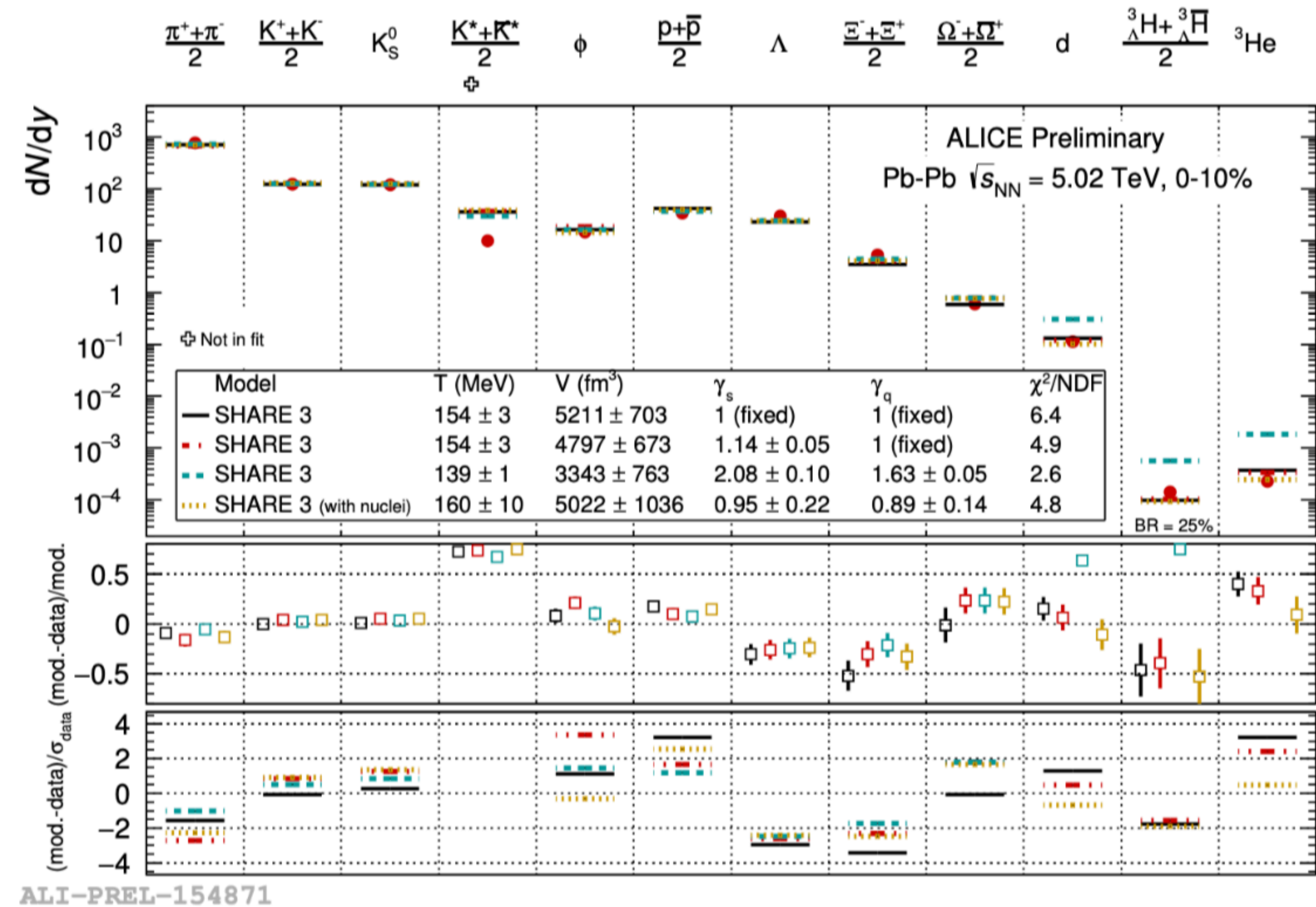


ALICE Coll., Physics Letters B734 (2014) 409–410, Physics Letters B 728 (2014) 216–227

Statistical Hadronization Models

- ❖ Hadron yields are well described by Statistical Hadronization Model (SHM)
- ❖ Indicates thermal + chemical equilibrium at freeze-out
- ❖ Central Pb–Pb collision (large system)
 - > “Grand canonical ensemble”
 - Strangeness conserved on average
 - Characterized by T, V, μ
- ❖ Extracted temperature: $T_{ch} \approx 155 \text{ MeV}$

$$N_i \propto \exp(-m/T_{ch})$$

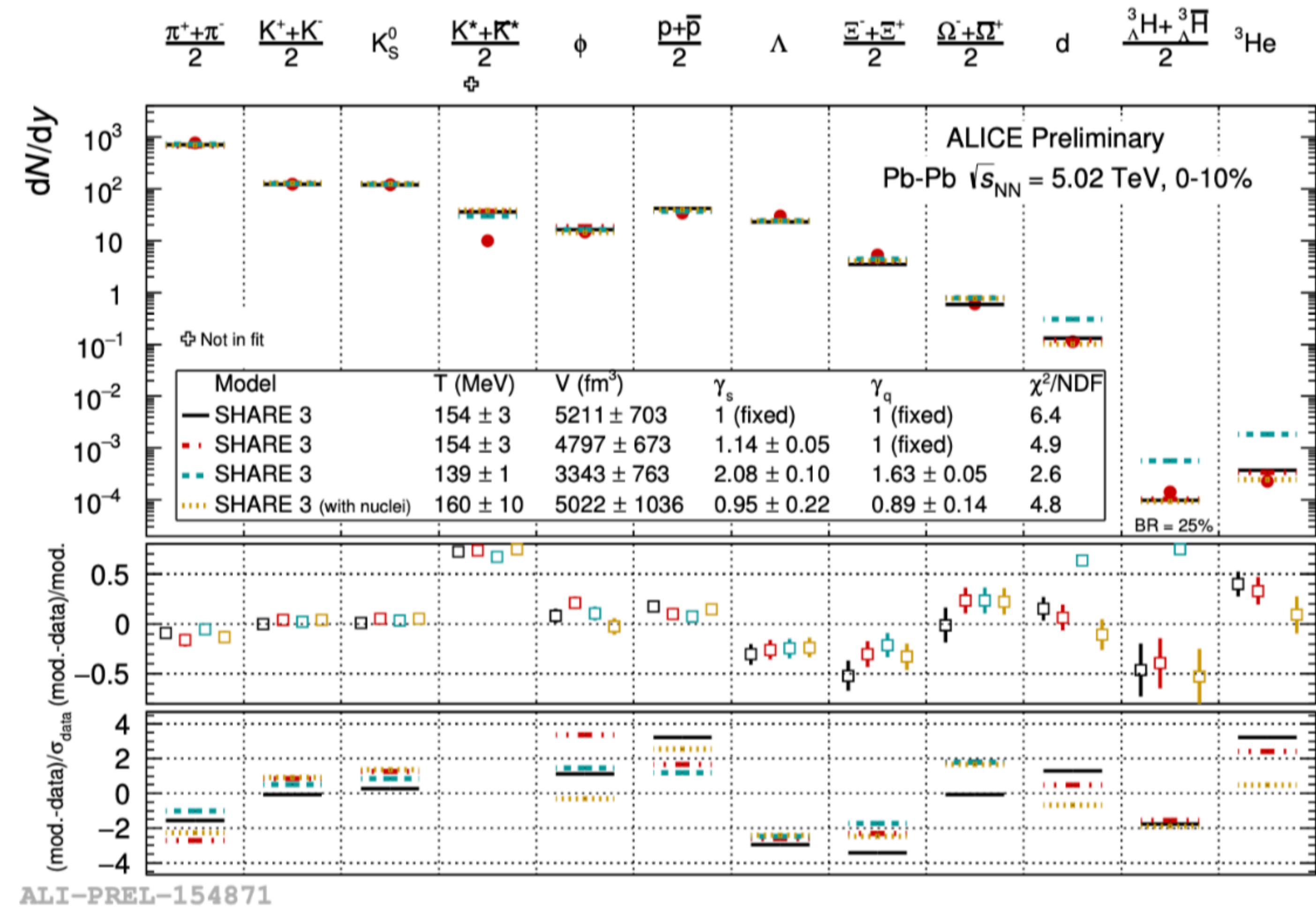


Statistical Hadronization Models

- ❖ Hadron yields are well described by Statistical Hadronization Model (SHM)
- ❖ Indicates thermal + chemical equilibrium at freeze-out
- ❖ Central Pb–Pb collision (large system)
 - > “Grand canonical ensemble”
 - Strangeness conserved on average
 - Characterized by T, V, μ
- ❖ Extracted temperature: $T_{ch} \approx 155 \text{ MeV}$

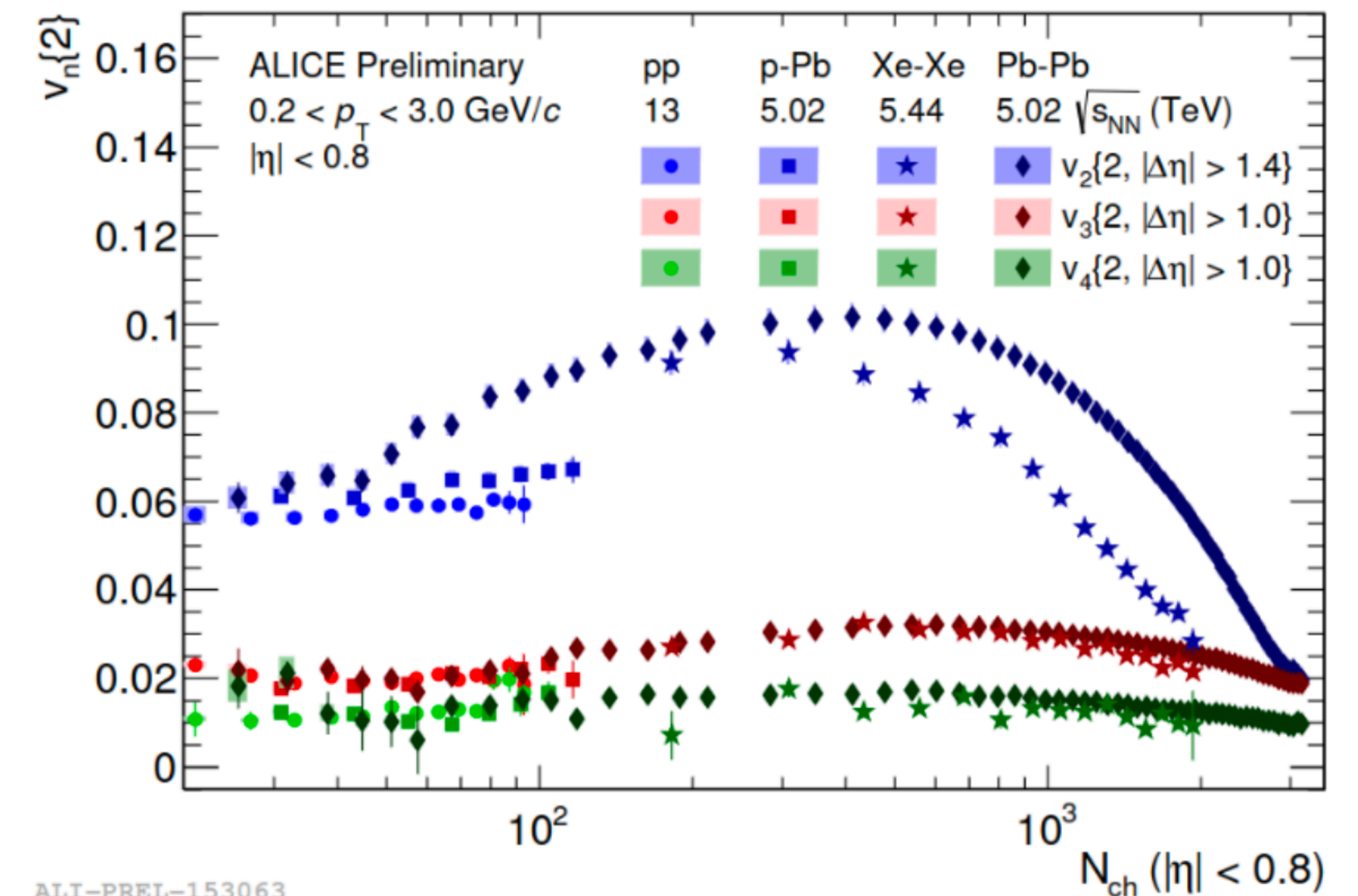
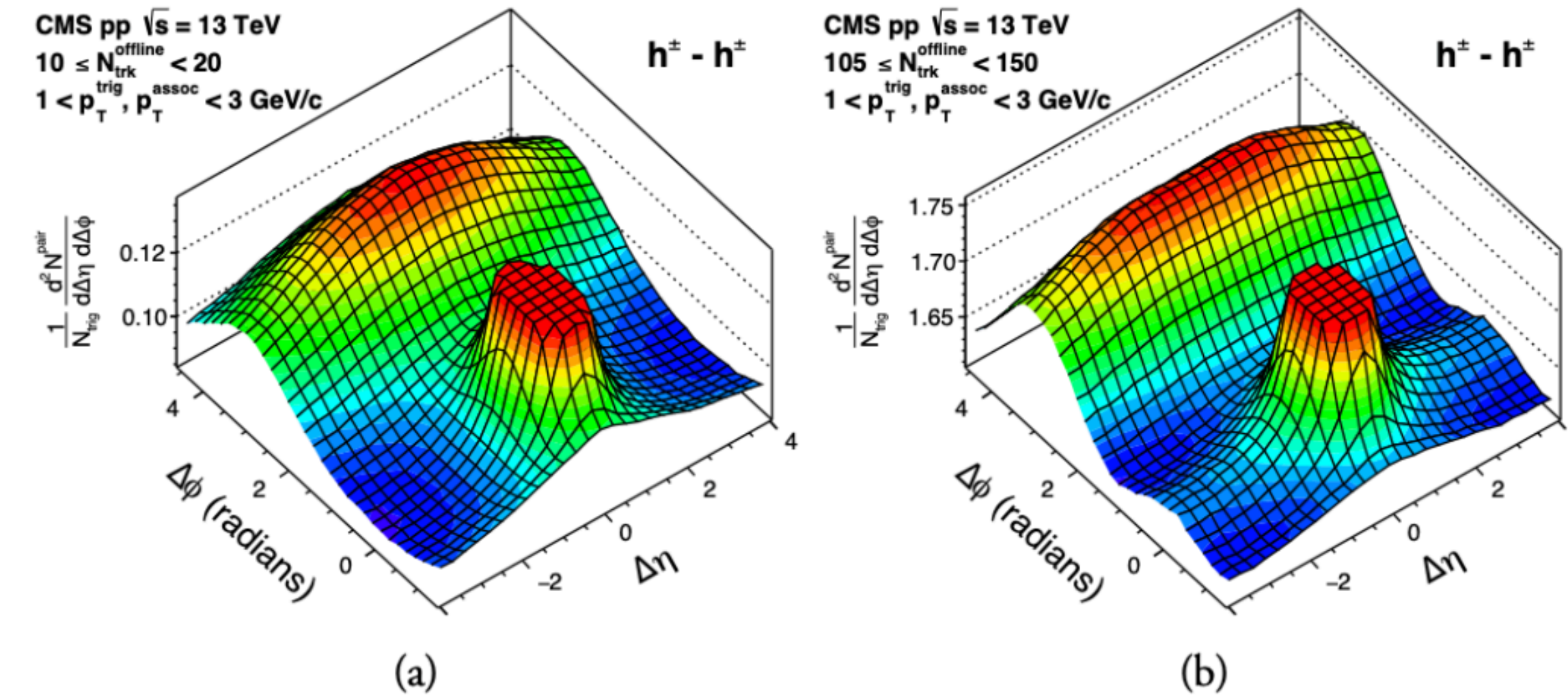
$N_i \propto \exp(-m/T_{ch})$

"Consistency check"



Small-System Surprise: High-Multiplicity pp and p-Pb

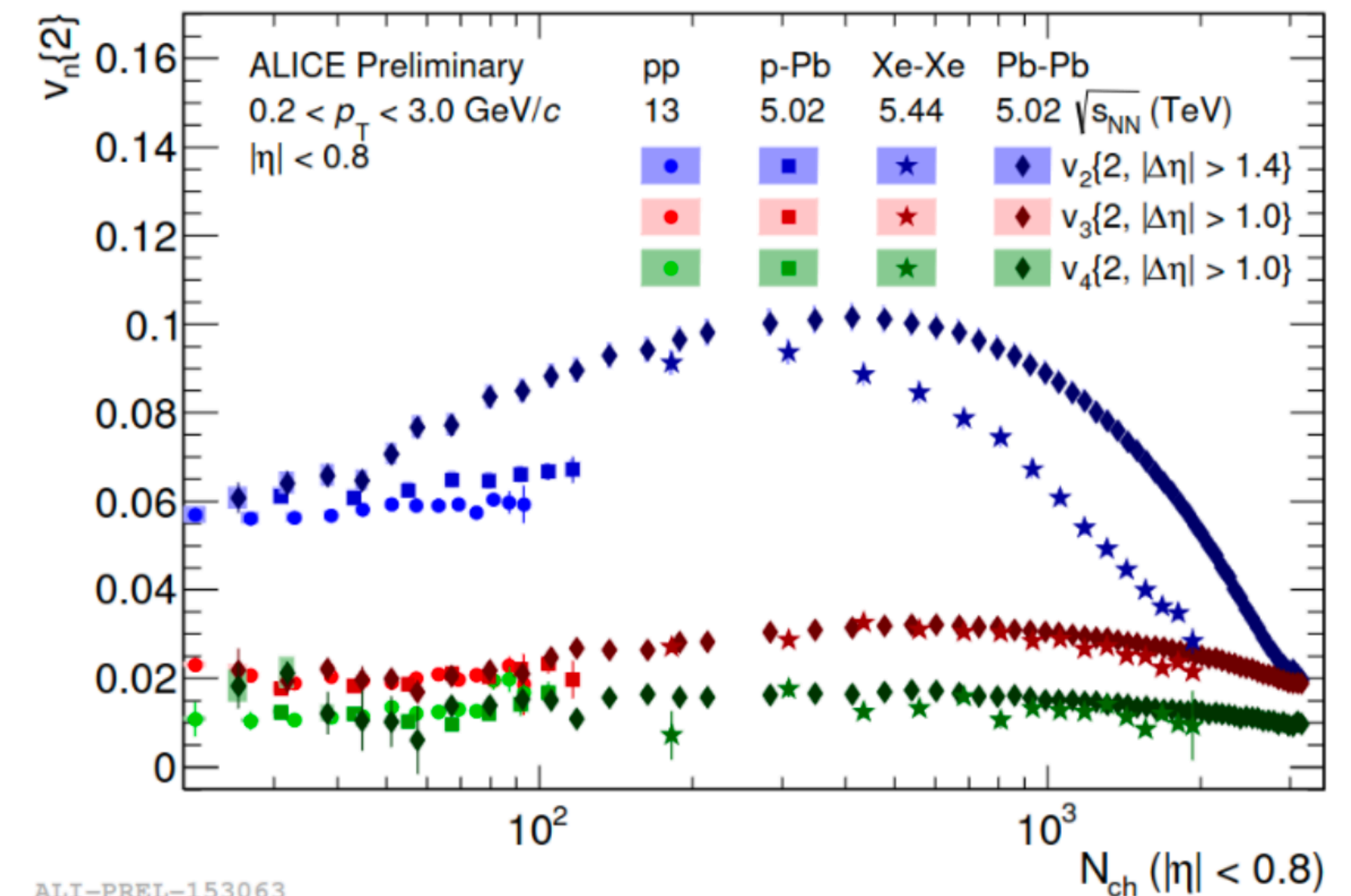
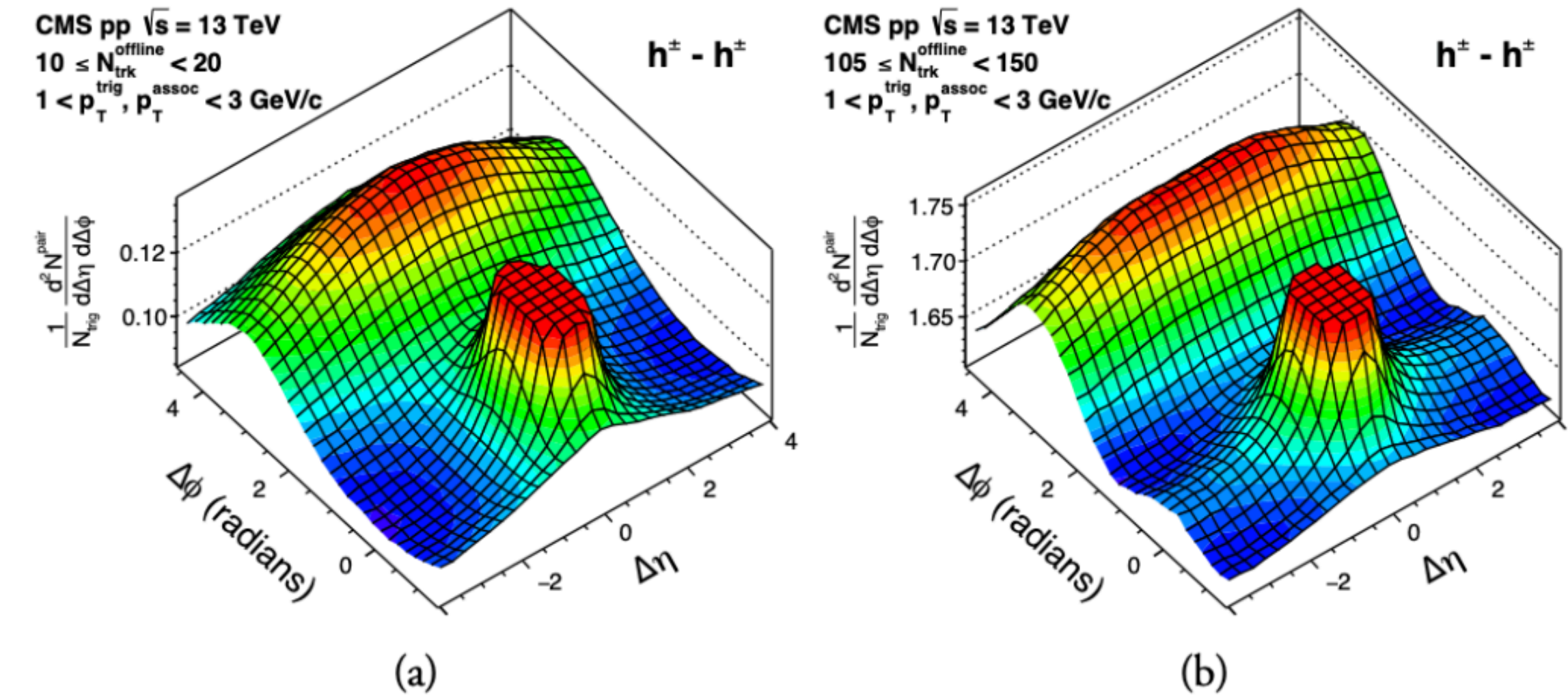
- ❖ MB (Minimum-bias) pp and p-Pb = reference systems
- ❖ Collective behavior was not expected in minimum-bias small systems.
- ❖ However, high-multiplicity events reveal:
 - near-side long-range ridge correlations
 - non-zero flow coefficients v_n
 - collectivity-like behavior resembling heavy-ion collisions



Small-System Surprise: High-Multiplicity pp and p-Pb

- ❖ MB (Minimum-bias) pp and p-Pb = reference systems
- ❖ Collective behavior was not expected in minimum-bias small systems.
- ❖ However, high-multiplicity events reveal:
 - near-side long-range ridge correlations
 - non-zero flow coefficients v_n
 - collectivity-like behavior resembling heavy-ion collisions

What about strangeness enhancement?



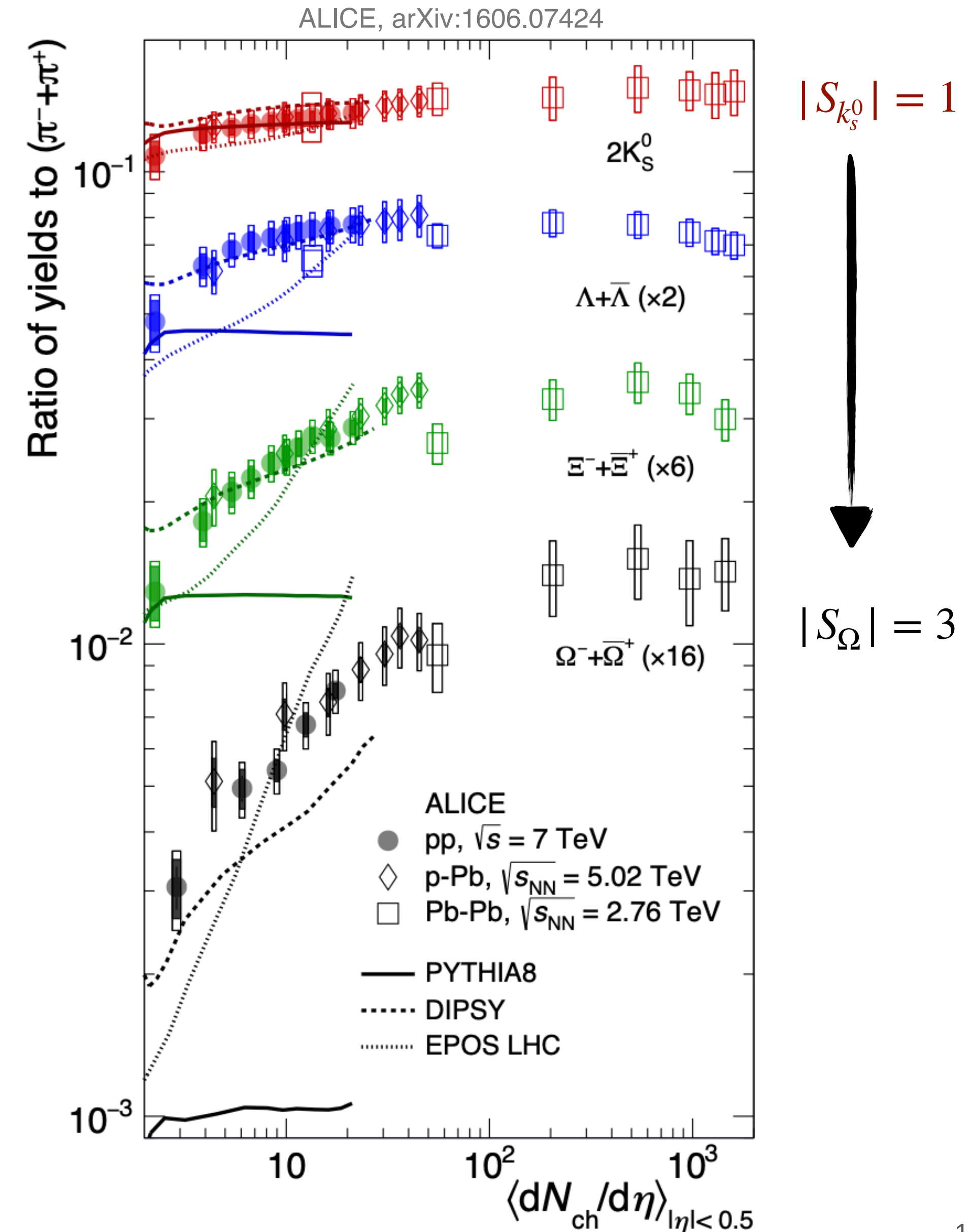
Strangeness Enhancement Across Collision Systems

❖ High-multiplicity pp and p-Pb collisions also exhibit strangeness enhancement!

❖ Enhancement increases with strangeness content:

$$K_S^0 \approx \frac{1}{2} (|d\bar{s}\rangle + |s\bar{d}\rangle) \quad \Lambda = |uds\rangle \quad \Xi^- = |dss\rangle \quad \Omega^- = |sss\rangle$$

❖ Scaling with multiplicity (not system type)



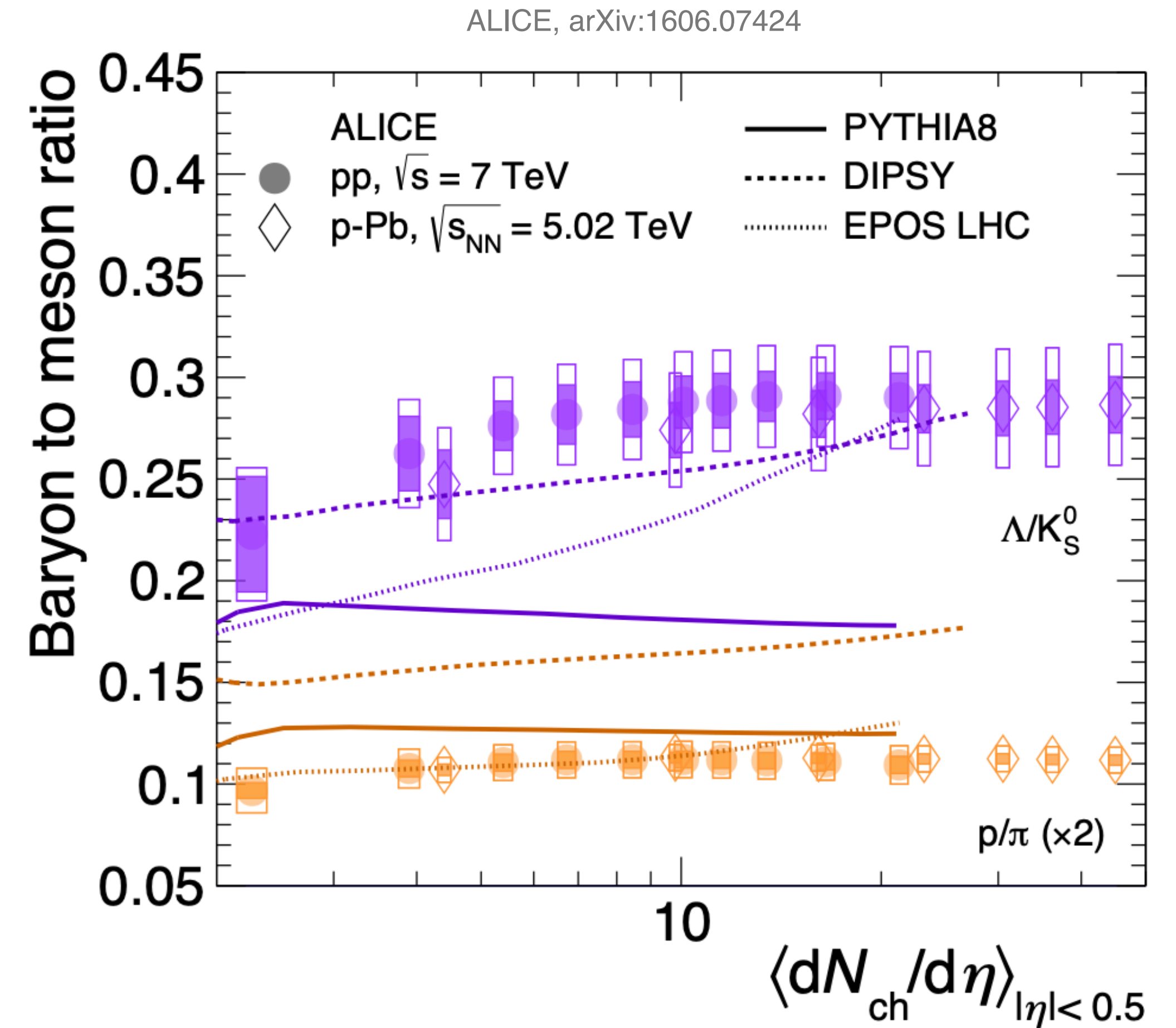
Not a Mass or Baryon-number Effect

$$\frac{\Lambda}{K_s^0} = \frac{\Lambda + \Lambda^-}{2k_s^0}$$

- ❖ Similar strange content
- ❖ Tests baryon-vs-meson effects

$$\frac{p}{\pi} = \frac{p + \bar{p}}{\pi^+ + \pi^-}$$

- ❖ Heavy non-strange baryon / light meson
- ❖ Nearly flat → not mass effect

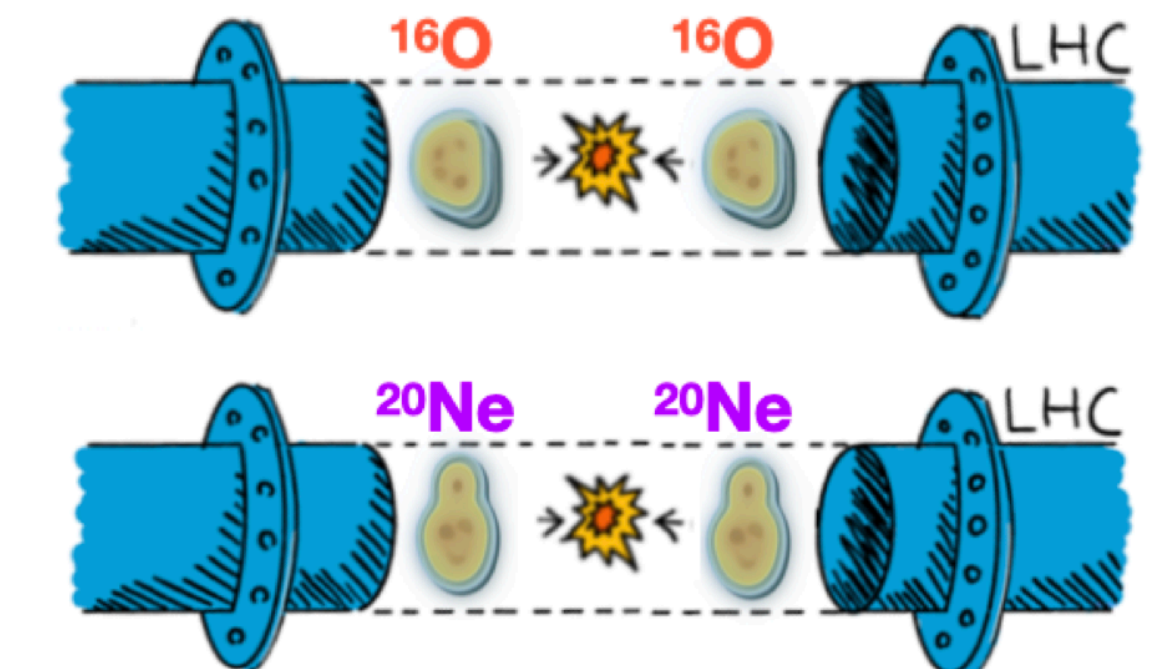
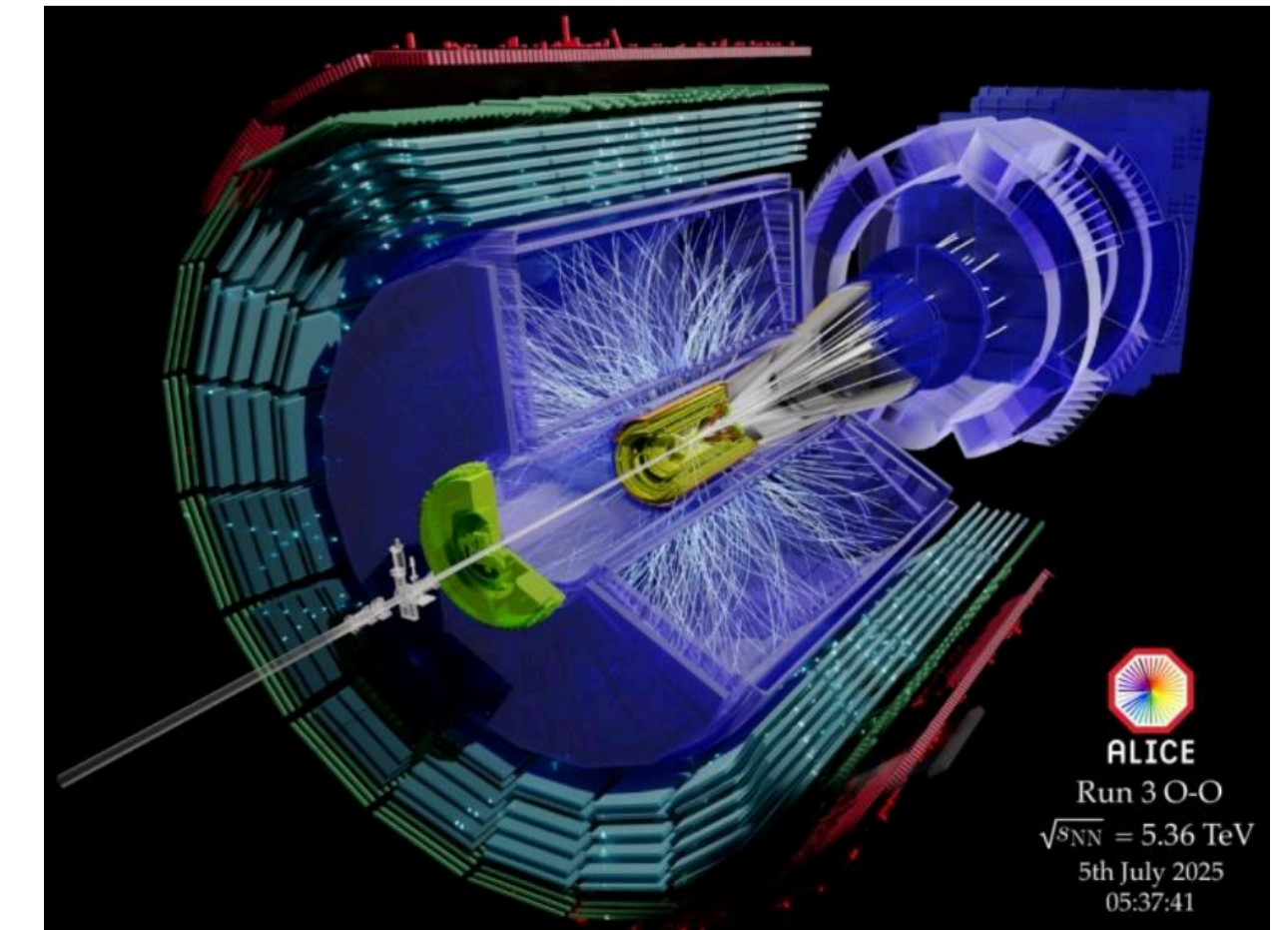


Strangeness Enhancement in OO

- ❖ First measurement of (multi-)strange hadron production in OO collisions at 5.36 TeV.
- ❖ OO results consistent with multiplicity scaling!
- ❖ Integrated yields show that OO bridges small and large collision systems.

In high multiplicity small pp, p-pb and intermediate OO collision events strangeness production reaches values similar to those observed in PbPb collisions, where a qgp is formed.

July 2025, First OO and Ne-Ne collisions at the LHC



Interpretation of Strangeness in Small Systems

Small systems → Exact local conservation

→ canonical suppression of strange yields

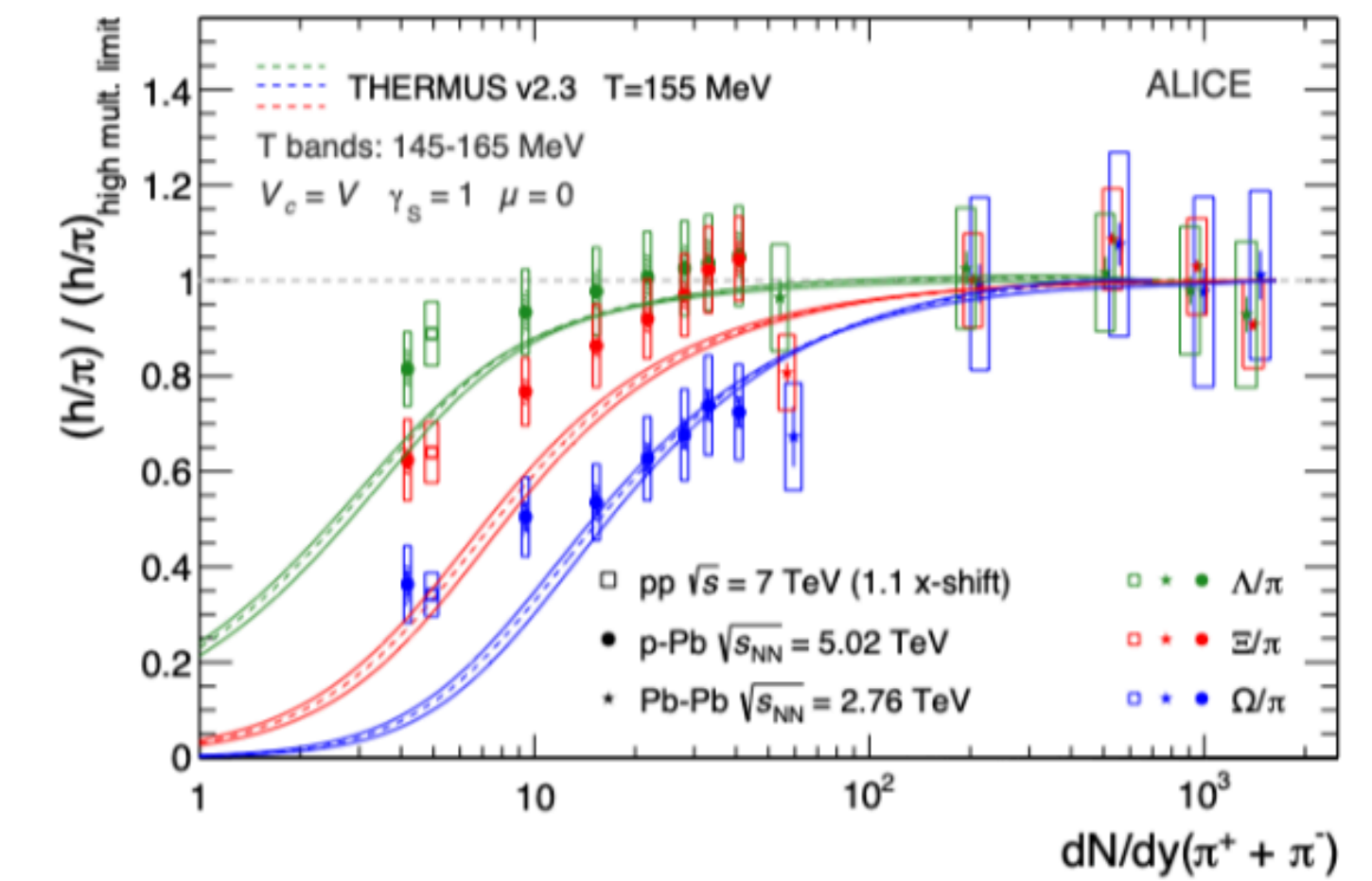
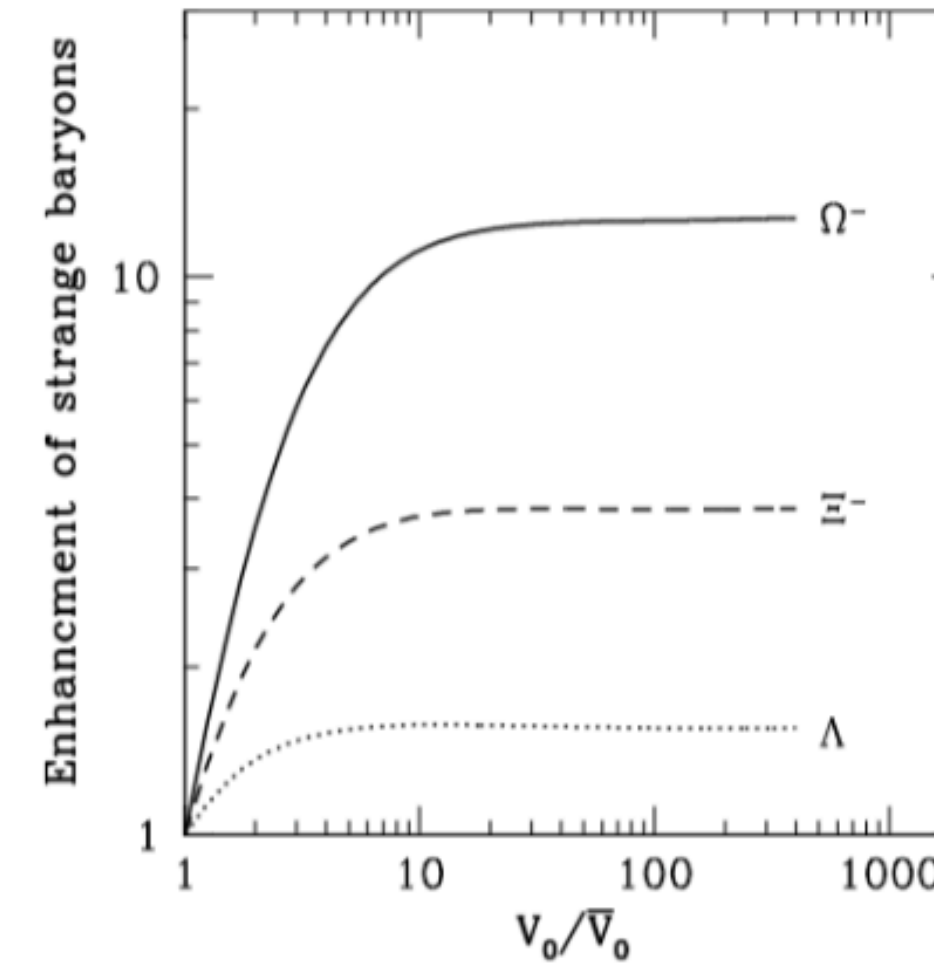
→ suppression decreases with multiplicity (effective volume)

Statistical Modelling

arXiv:hep-ph/0006024

❖ Canonical suppression (statistical model extension)

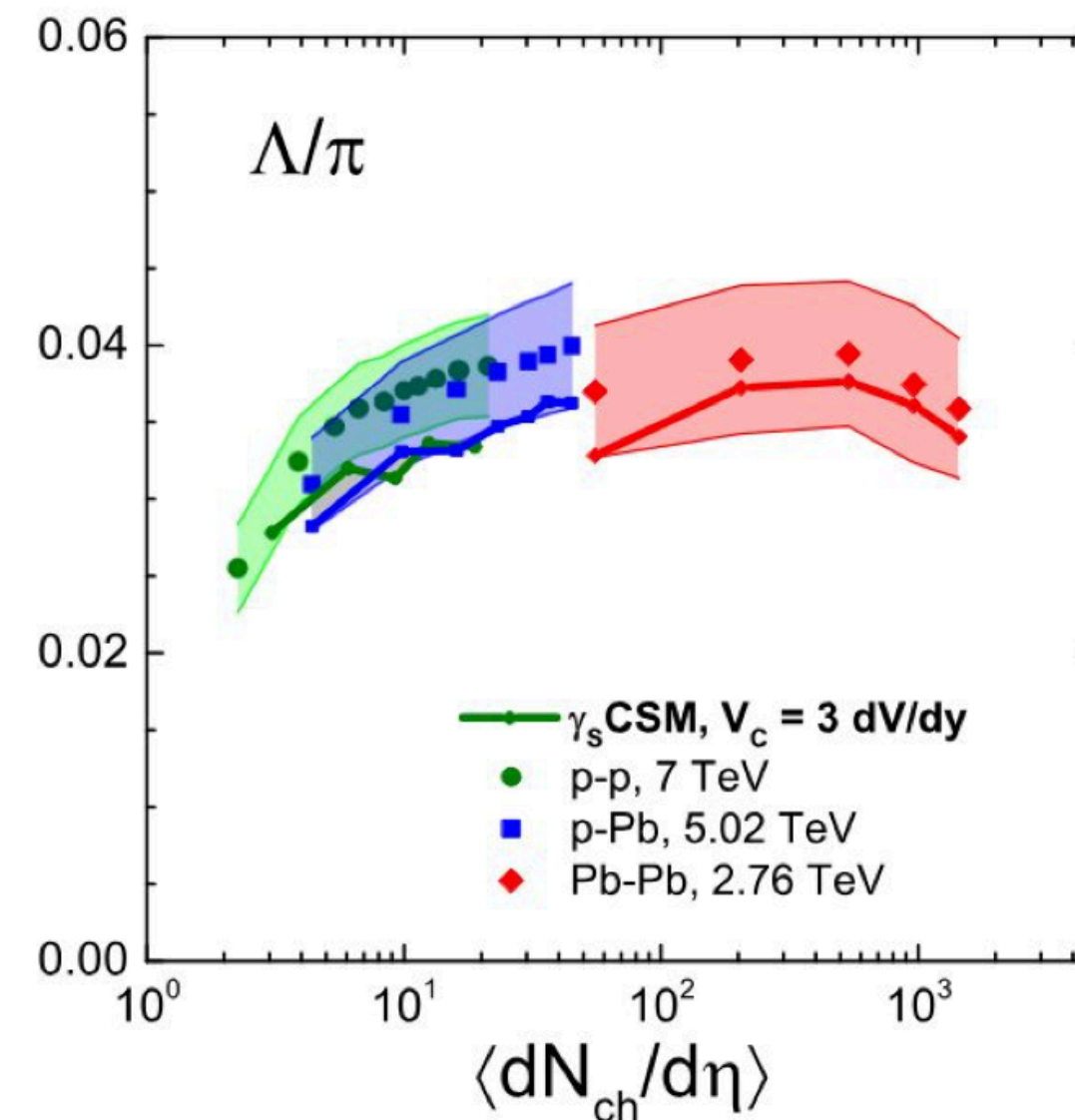
- Extend SHM to small systems
- Exact conservation enforced in a finite volume (V_c)
 - Reduced phase space
 - suppressed strange yields



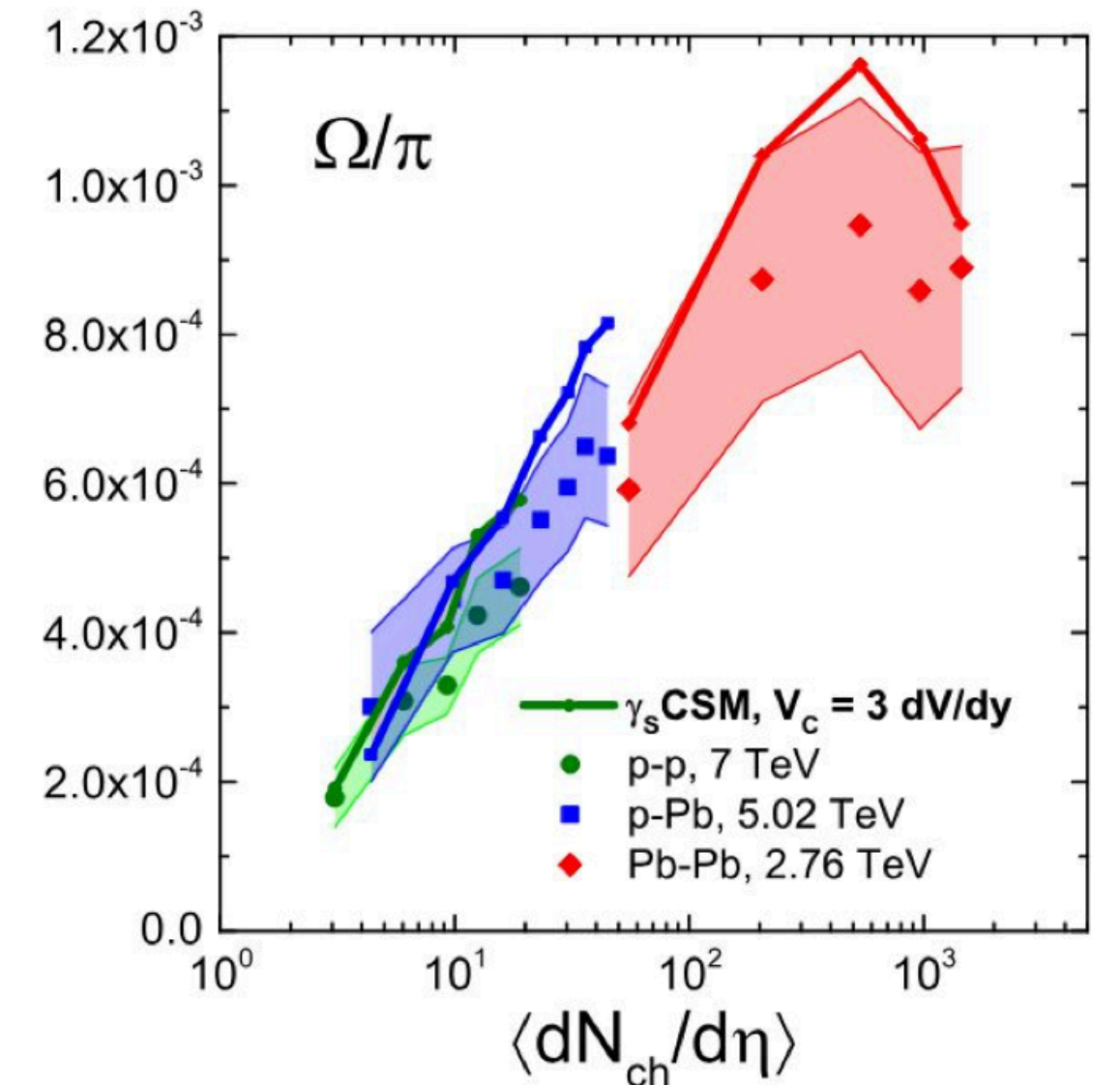
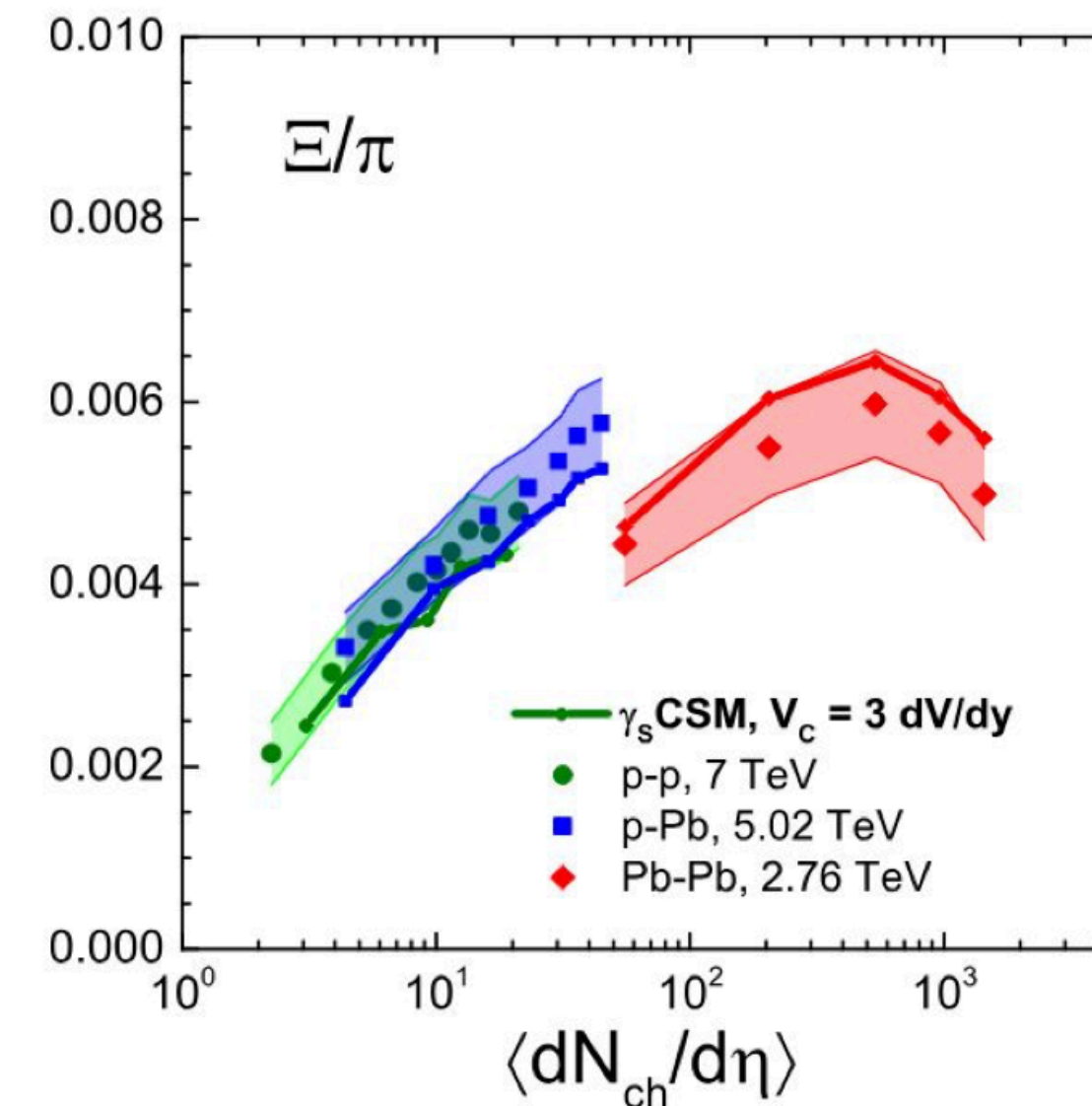
❖ Strangeness saturation (γ_s)

- $\gamma_s < 1$ → incomplete equilibration (small systems)
- $\gamma_s \rightarrow 1$ → approach equilibrium (large systems)

V. Vovchenko et al., Phys. Rev. C 100, 054906 (2019)



J. Cleymans et al, Phys. Rev. C 103, 014904 (2021)



Increasing multiplicity $\approx$ increasing effective volume $\rightarrow$ system transitions from canonical $\rightarrow$ grand canonical

Modeling Strangeness in Small Systems

❖ Other approaches focus on hadronization dynamics:

- Rope fragmentation: overlapping strings enhance ss^-
- Color reconnection: modified hadronization topology
- Core-corona models (EPOS): dense thermal core + dilute string-like corona

No single mechanism explains all data!!

Multiple mechanisms can reproduce the observed trends or different parts of it.

—> No unique explanation of strangeness enhancement in small systems.

What Drives Strangeness Enhancement?

❖ Microscopic origin of Strangeness Enhancement is still debated, especially in small systems.

What drives strange hadron production ?!

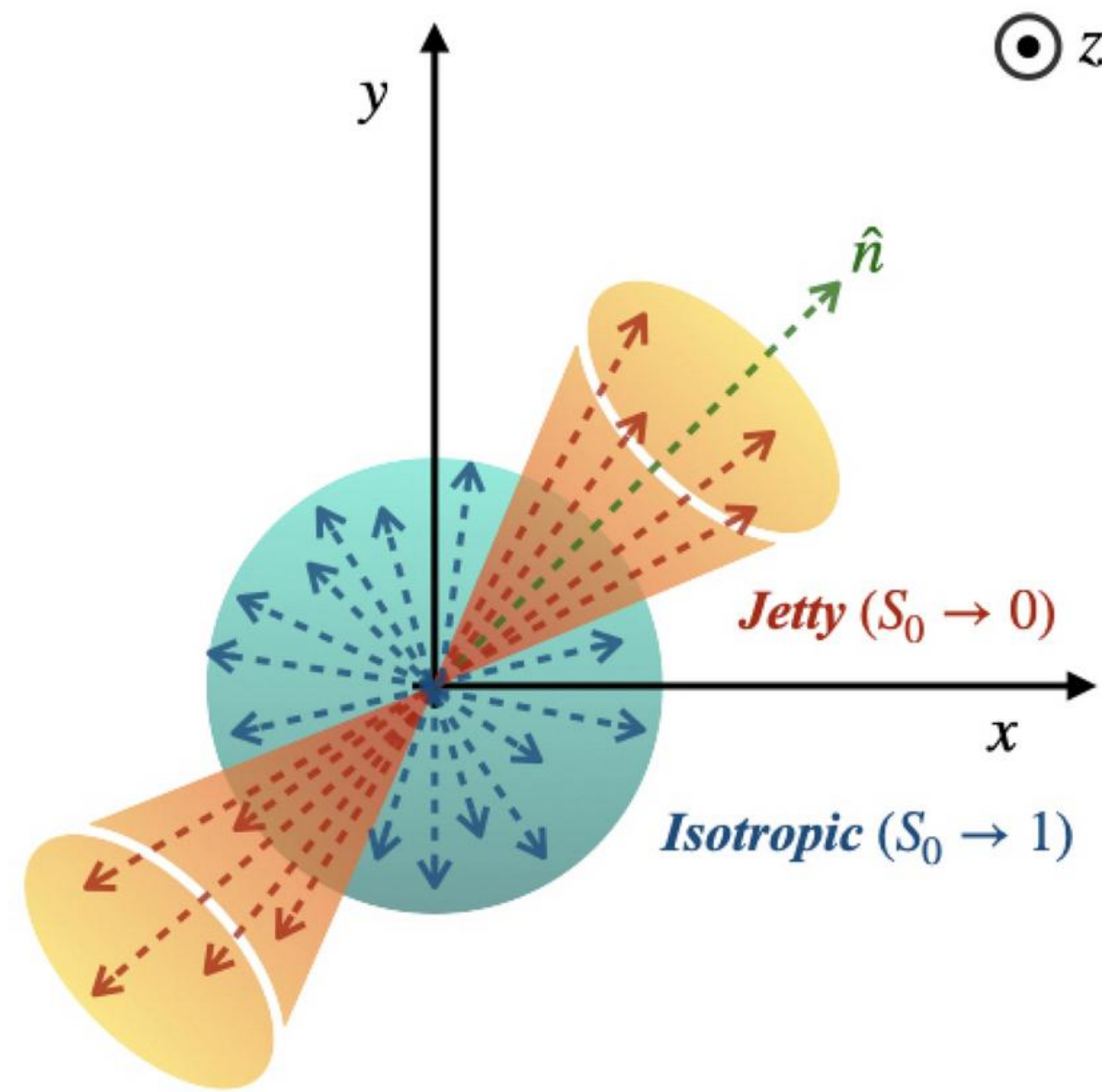
Beyond Multiplicity: Event Topology

❖ Event-shape engineering isolates topologies

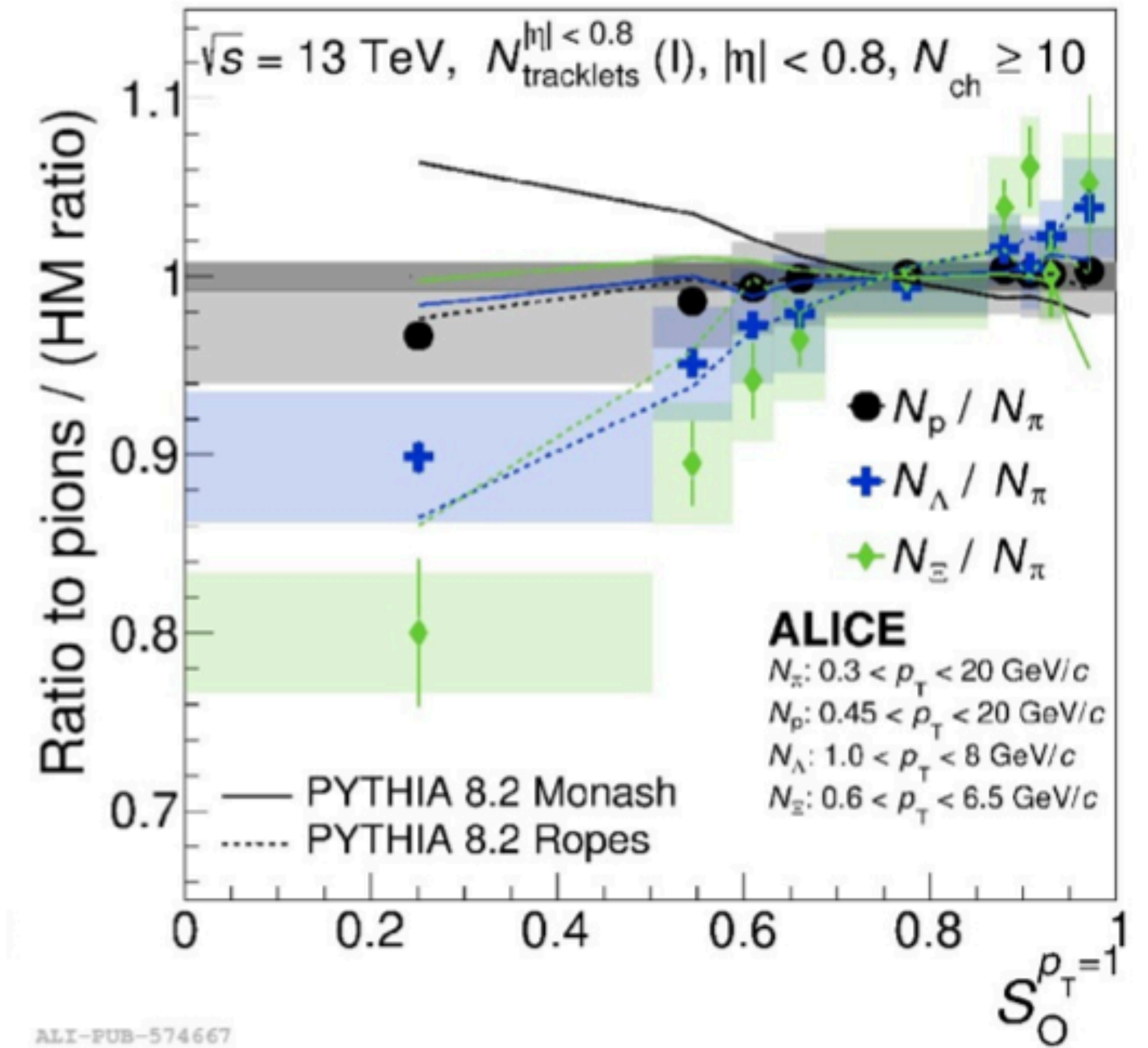
[ALICE spherocity JHEP 2024](#), ALICE reported in High Multiplicity pp:

- Isotropic soft events → stronger strange particle production
- Jet-like events → suppressed strangeness

∴ Event Topology Matters



Prasad, et al. *Sci Rep* **12**, 3917 (2022).



$$S_0^{p_T=1} = \frac{\pi^2}{4} \min_{\hat{n}} \left(\frac{\sum_i |\hat{p}_{Ti} \times \hat{n}|}{N_{\text{trks}}} \right)^2$$

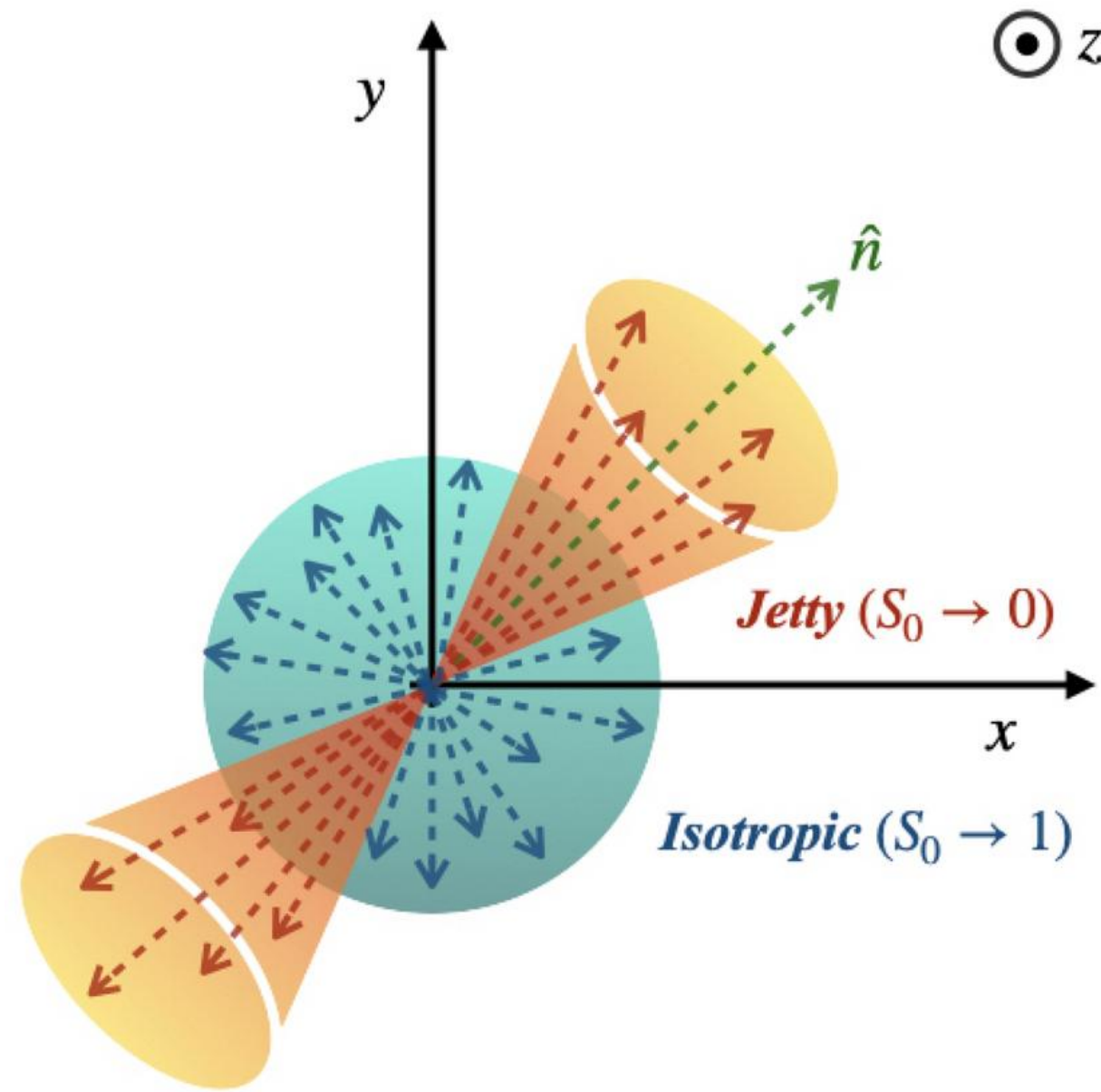
Beyond Multiplicity: Event Topology

❖ Event-shape engineering isolates topologies

[ALICE spherocity JHEP 2024](#), ALICE reported in High Multiplicity pp:

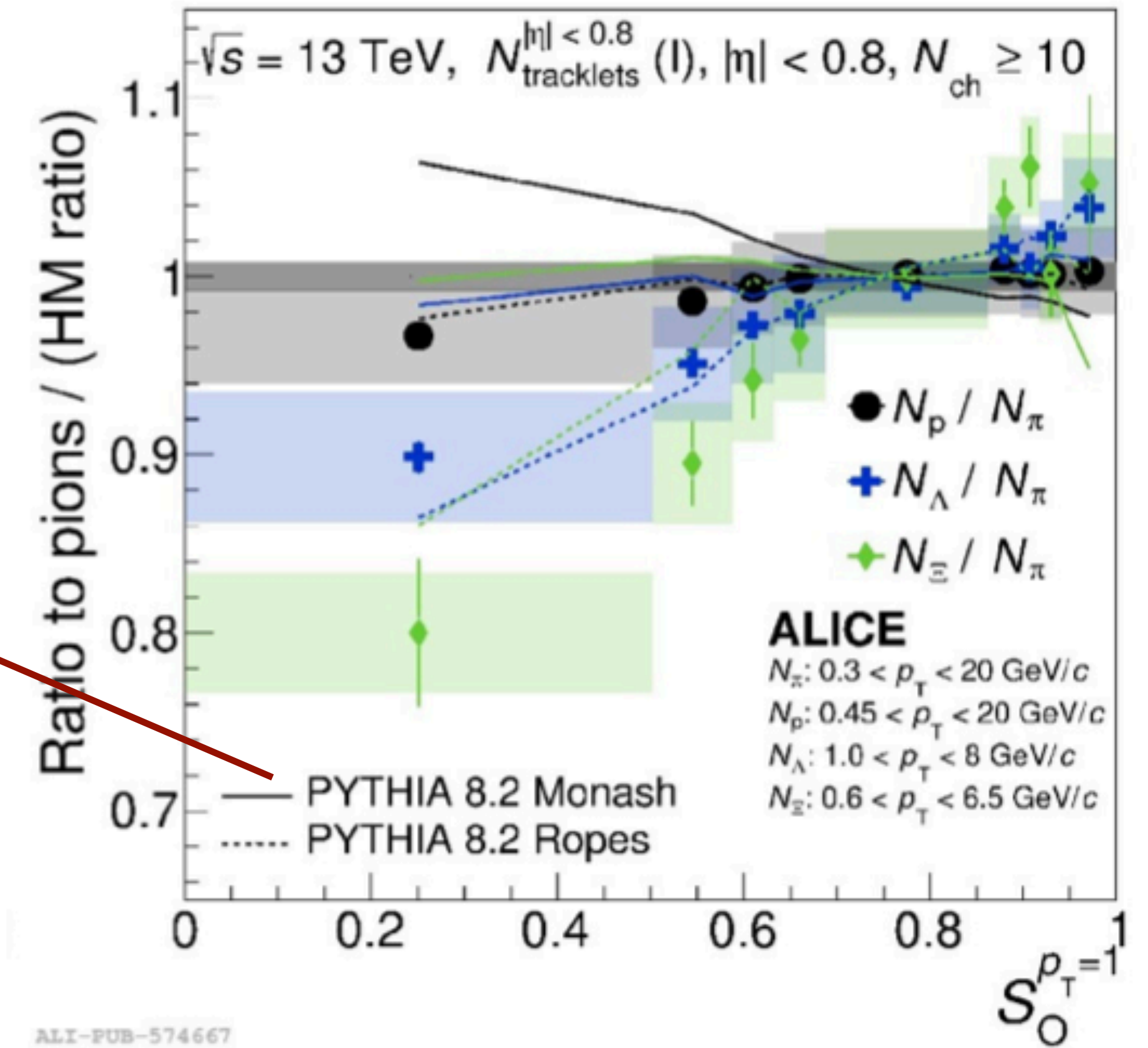
- Isotropic soft events → stronger strange particle production
- Jet-like events → suppressed strangeness

∴ Event Topology Matters



Prasad, et al. *Sci Rep* **12**, 3917 (2022).

- Pythia 8.2 Ropes:
reproduce the general trend, it fails to capture the strangeness hierarchy
- Pythia 8.2 Monash:
underestimates all ratios



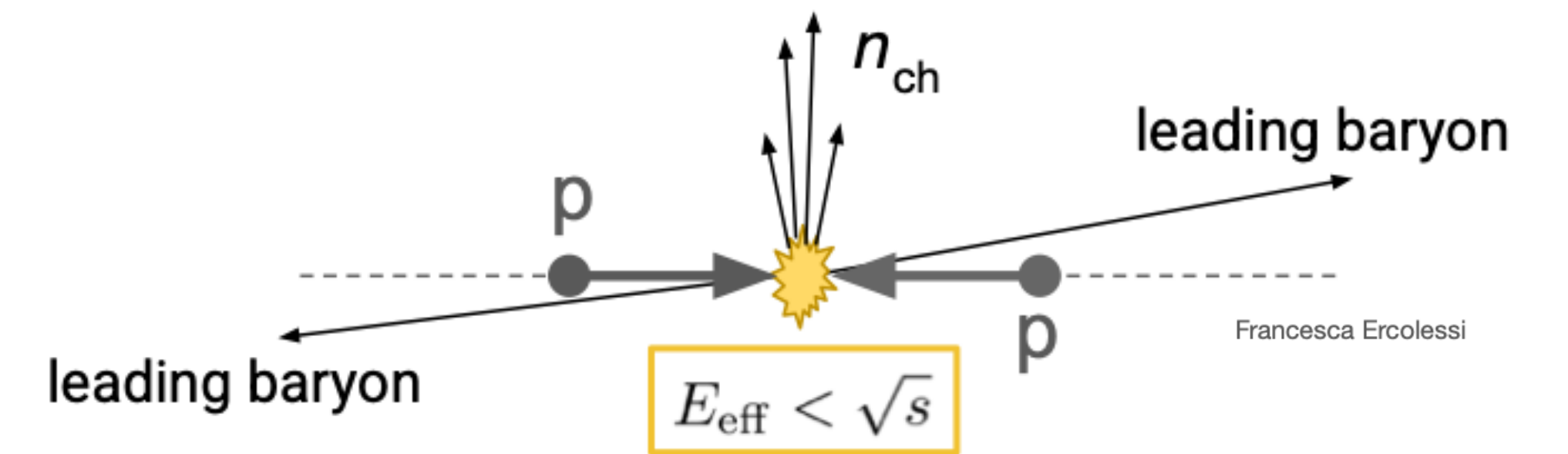
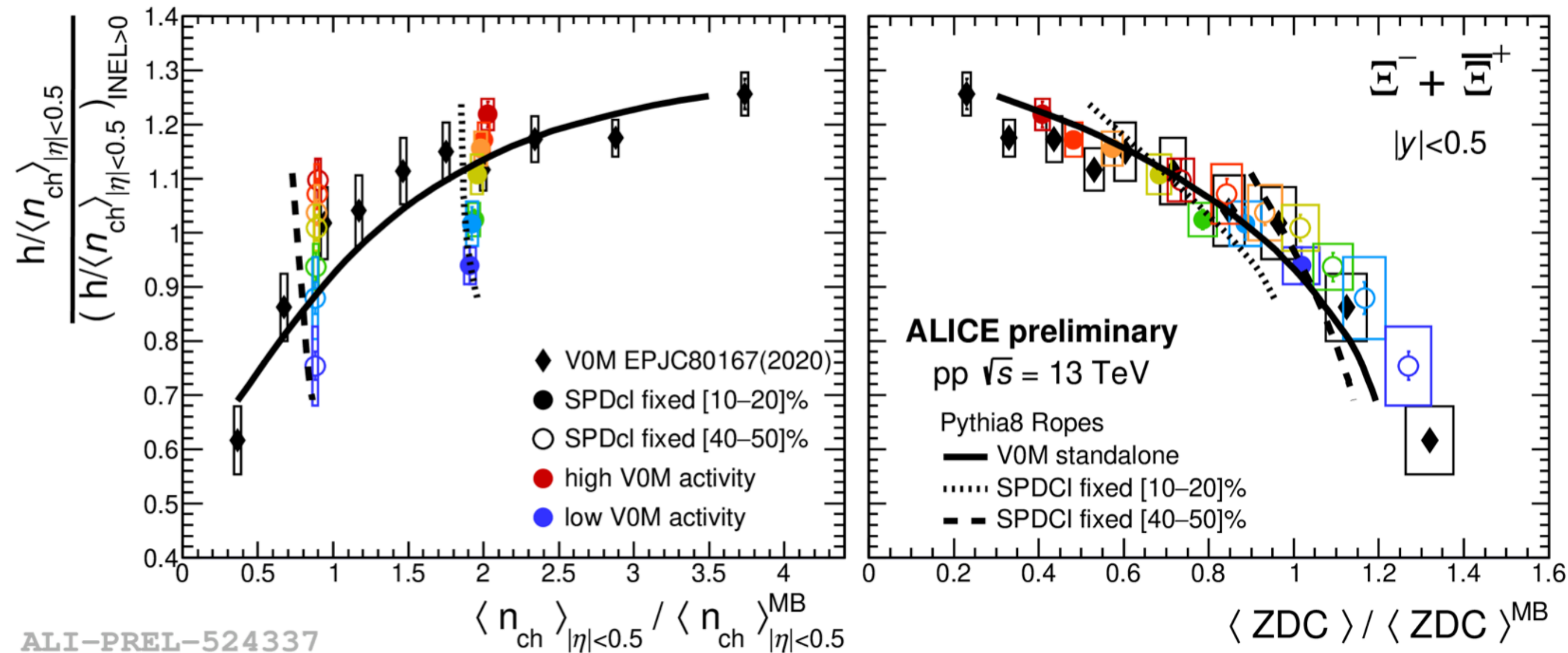
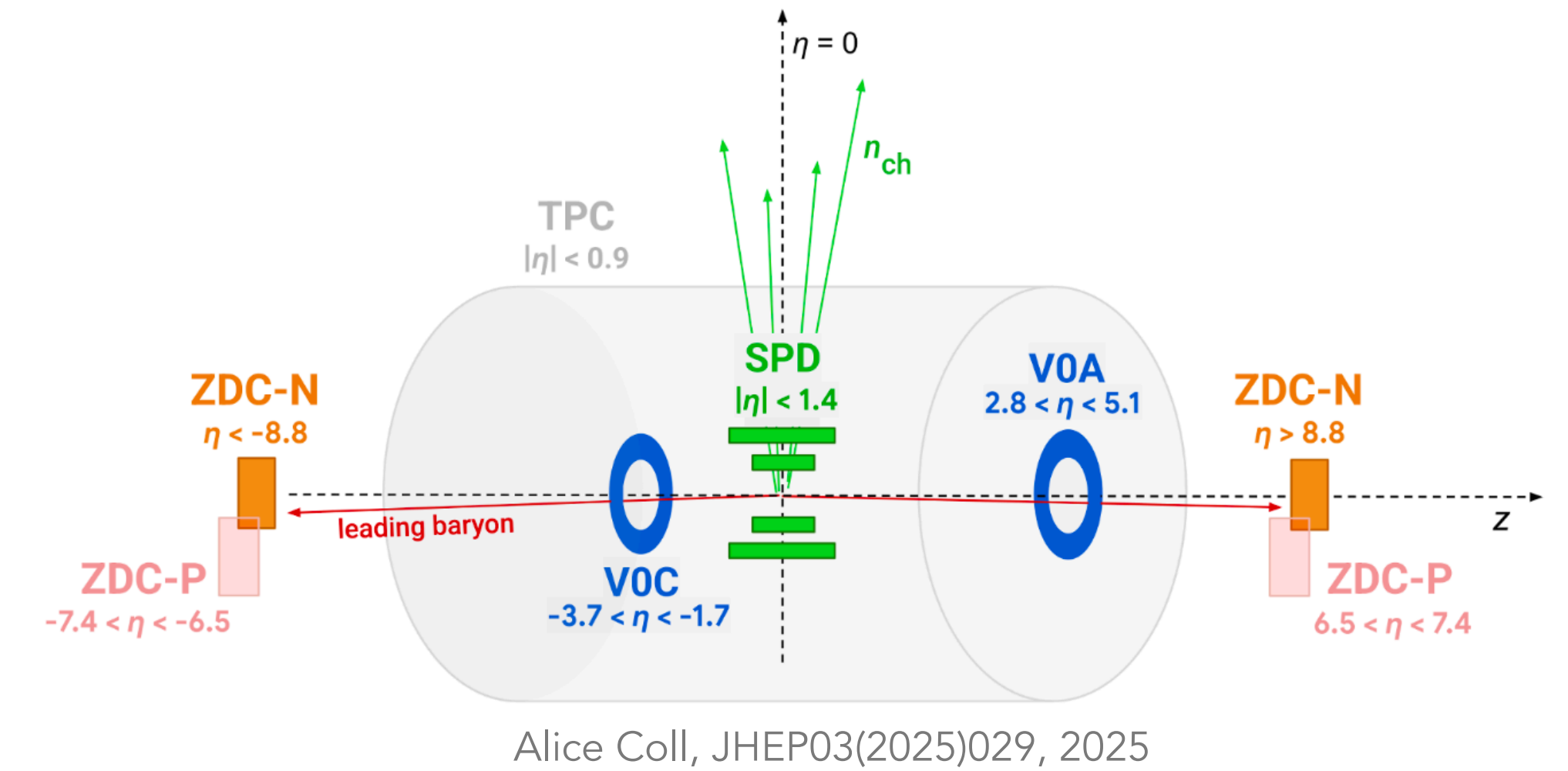
$$S_0^{p_T=1} = \frac{\pi^2}{4} \min_{\hat{n}} \left(\frac{\sum_i |\hat{p}_{Ti} \times \hat{n}|}{N_{\text{trks}}} \right)^2$$

Beyond Multiplicity: Effective Energy

❖ Effective energy

[ALICE Collaboration, JHEP 2025, 29 \(2025\):](#) "First observation of strange baryon enhancement with effective energy in pp collisions at the LHC"

- Recent forward energy studies show multiplicity alone may be insufficient.
- Effective energy $\equiv$ energy available for particle production in the collision.



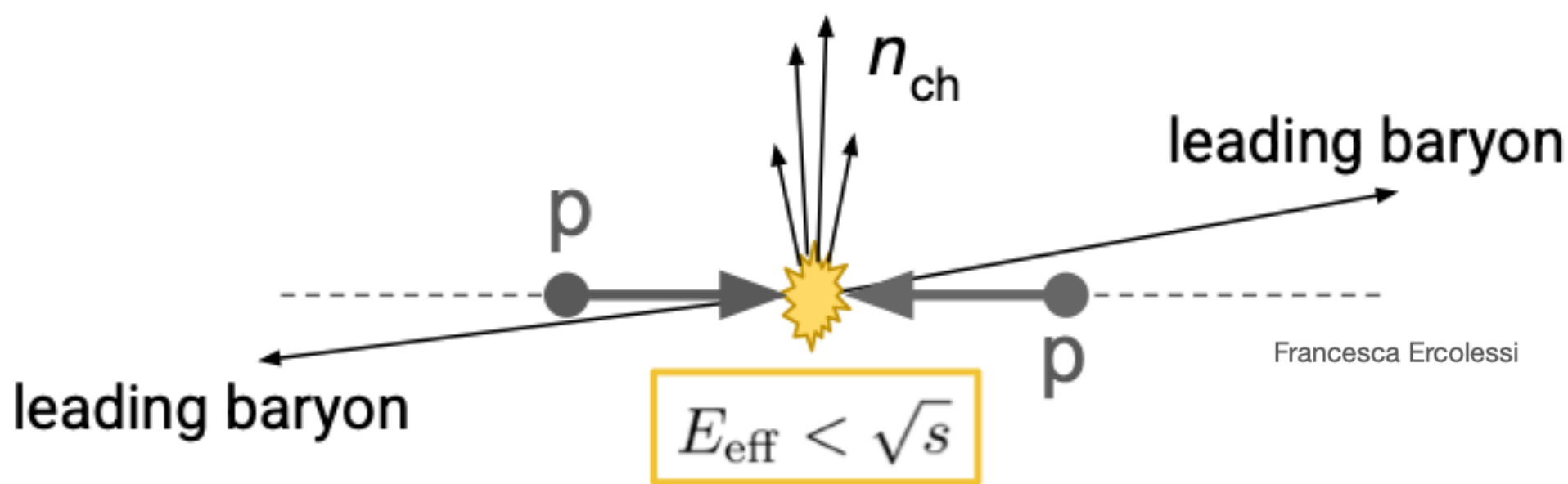
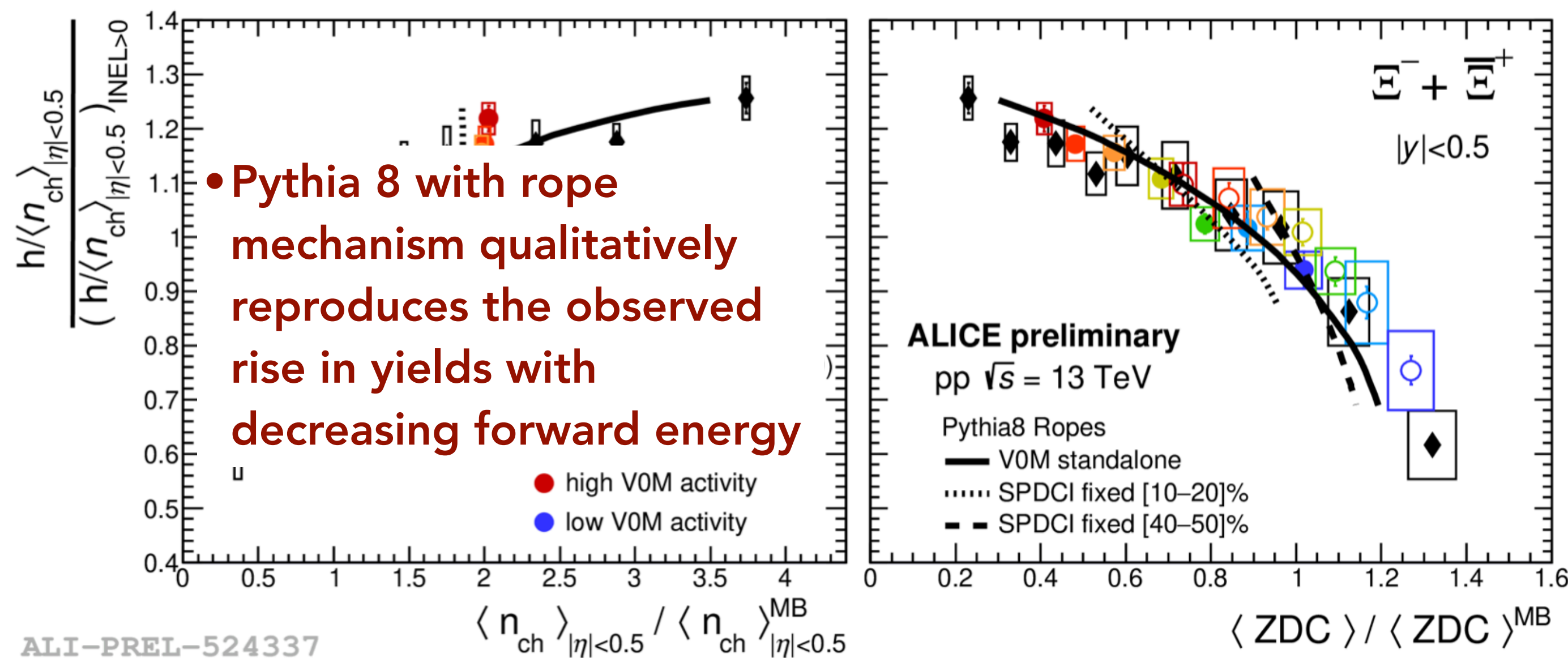
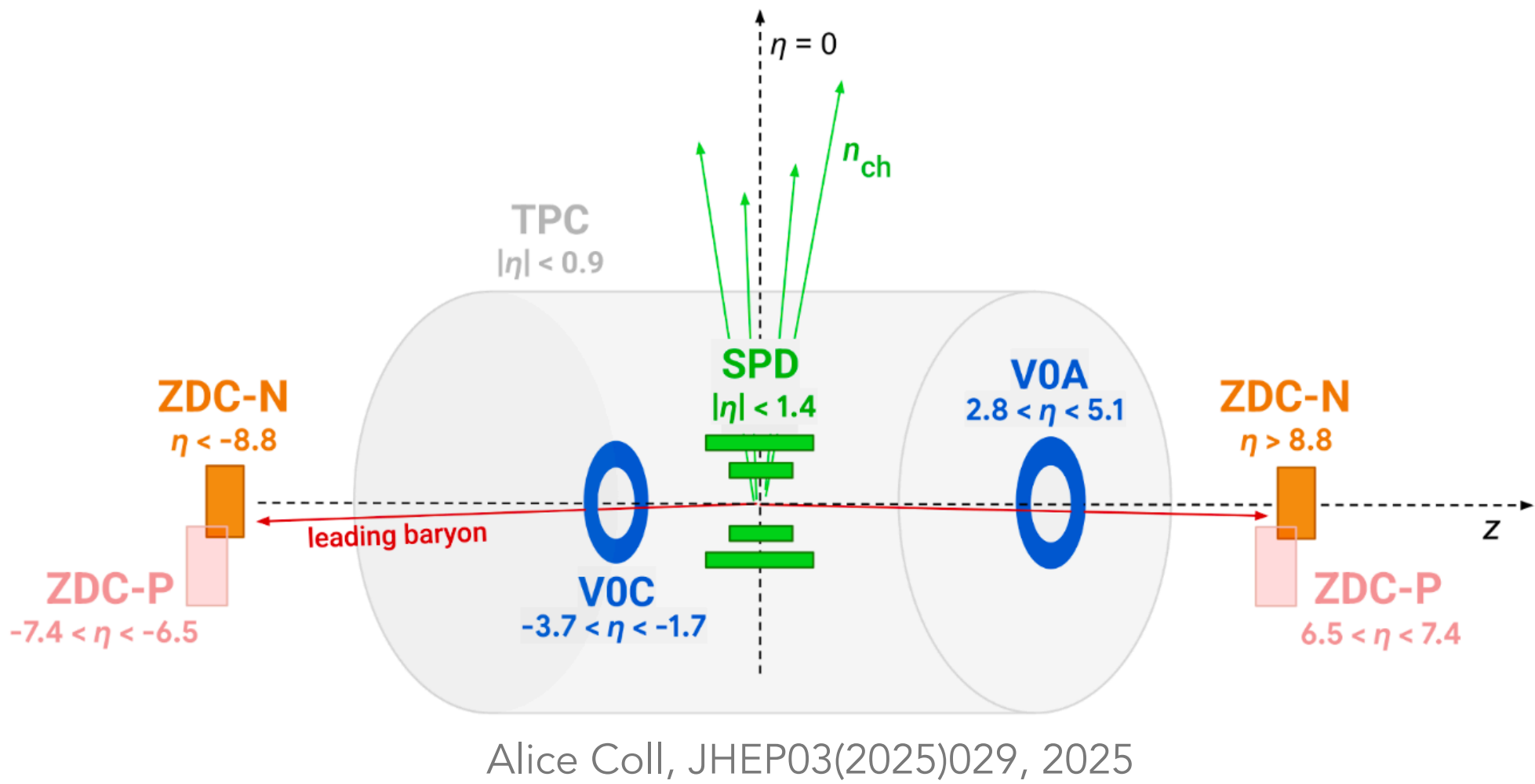
$$E_{eff} = \sqrt{s} - E_{leading} \approx \sqrt{s} - E_{forward}$$

Beyond Multiplicity: Effective Energy

❖ Effective energy

[ALICE Collaboration, JHEP 2025, 29 \(2025\):](#) “First observation of strange baryon enhancement with effective energy in pp collisions at the LHC”

- Recent forward energy studies show multiplicity alone may be insufficient.
- Effective energy $\equiv$ energy available for particle production in the collision.

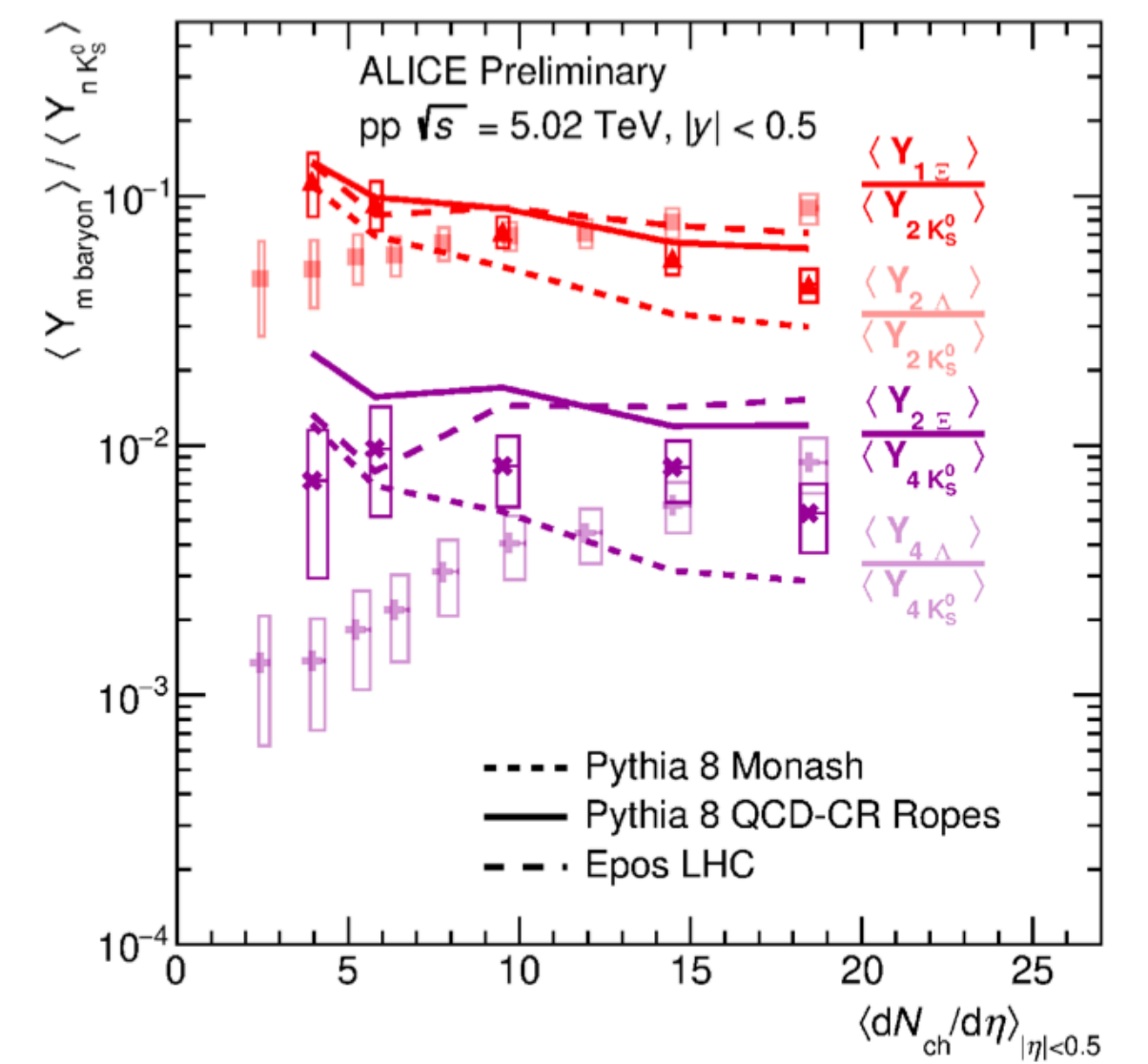
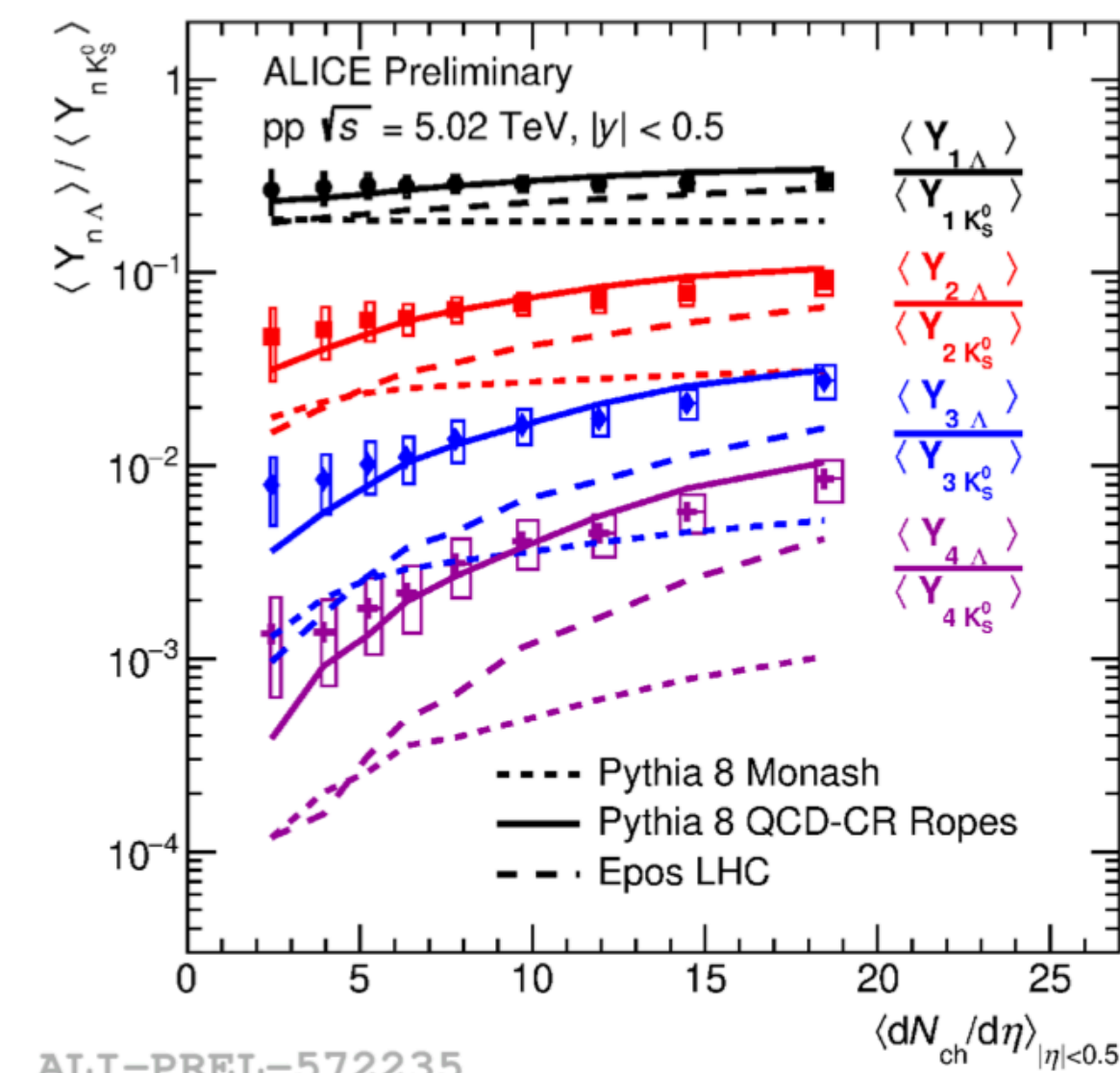
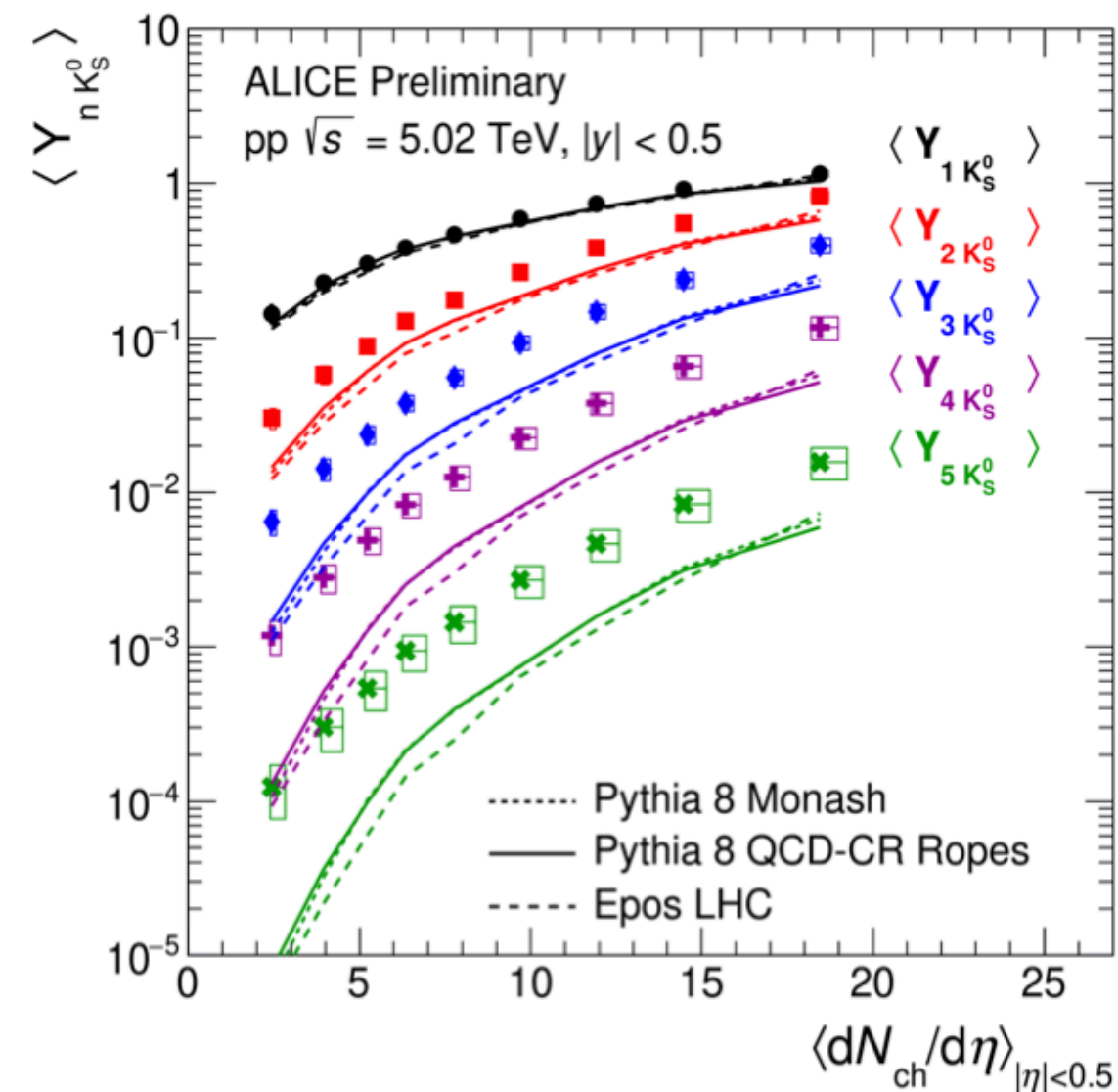
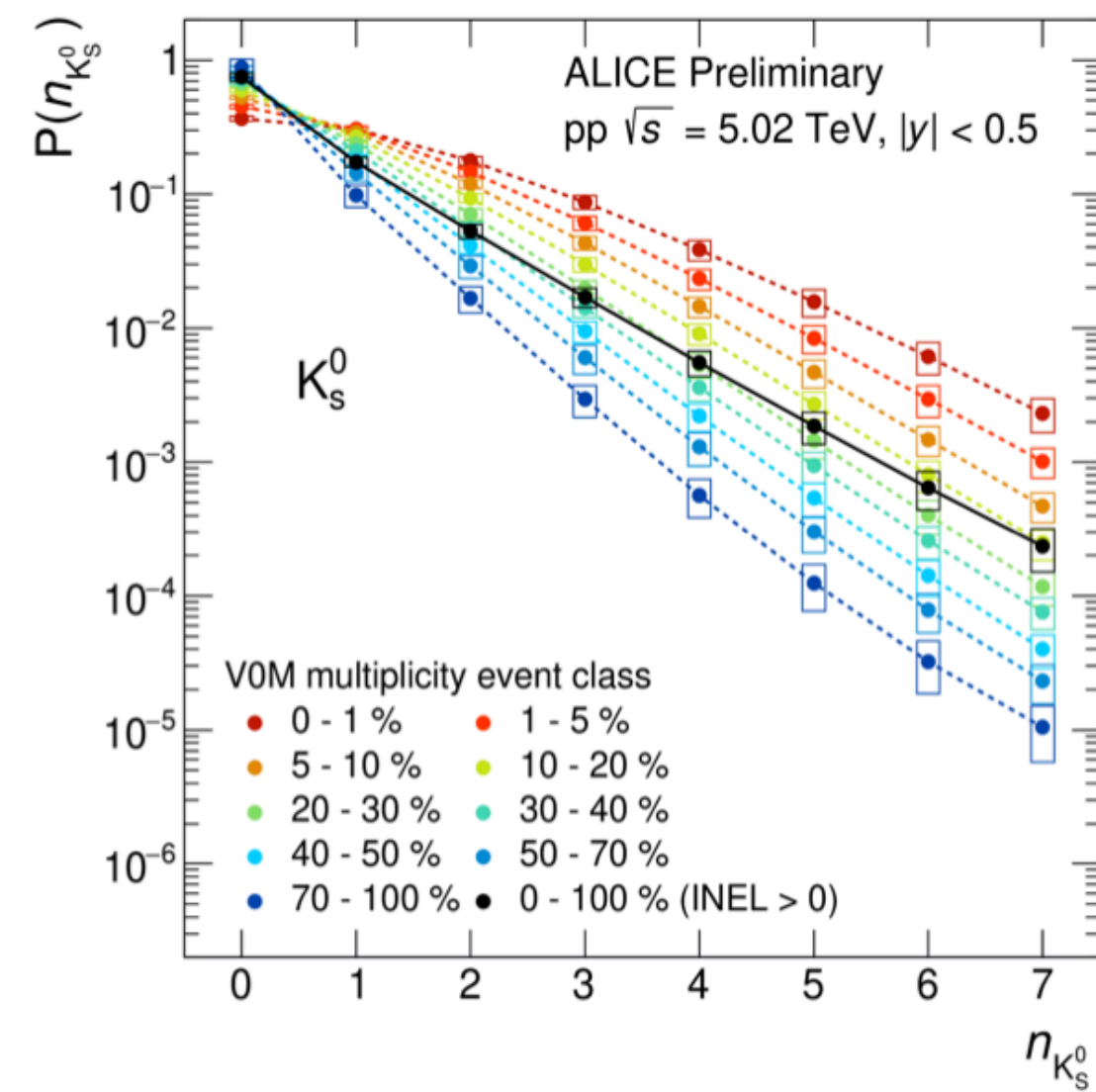


$$E_{eff} = \sqrt{s} - E_{leading} \approx \sqrt{s} - E_{forward}$$

Strange Particle Multiplicity Distributions

- ❖ New experimental observable proposed : strange particle multiplicity distribution ([arXiv:2511.10413](https://arxiv.org/abs/2511.10413))
- ❖ First measurement of the probability to produce n particles of a given species $P(n_S)$ per events
- ❖ Event-by-event strange-particle multiplicity distributions probe fluctuations and rare production

$$\langle Y_{nS} \rangle = \sum_{i=n}^{\infty} \frac{i!}{n!(i-n)!} \cdot P(i_S).$$



Summary & Outlook

- ❖ Strangeness has evolved from a QGP signature to a broader probe of QCD dynamics and hadronization.
- ❖ Strangeness enhancement is not exclusive to heavy-ion systems!
- ❖ Smooth scaling with multiplicity across systems
- ❖ Enhancement increases with strange-quark content
- ❖ No single mechanism explains all data → interplay of effects
- ❖ Further model developments and new observables are needed to reach a complete microscopic understanding of strangeness enhancement, especially at the LHC.

References

1. Rafelski & Müller, Strangeness Production in the Quark-Gluon Plasma:
<https://cds.cern.ch/record/147788/files/198311105.pdf>
2. ALICE Collaboration, Enhanced production of multi-strange hadrons in high-multiplicity proton–proton collisions. <https://www.nature.com/articles/nphys4111>
3. ALICE Collaboration, Recent results on strangeness enhancement in small collision systems with ALICE. <https://arxiv.org/abs/2504.02527> and references within it.
4. Rafelski, *Strangeness in Quark–Gluon Plasma – 1982*
https://link.springer.com/chapter/10.1007/978-3-319-17545-4_31
5. Multi-strange baryon production at mid-rapidity in Pb–Pb collisions at $\sqrt{s_{NN}} = 2.76 \text{ TeV}$
<https://arxiv.org/pdf/1307.5543>

Backup

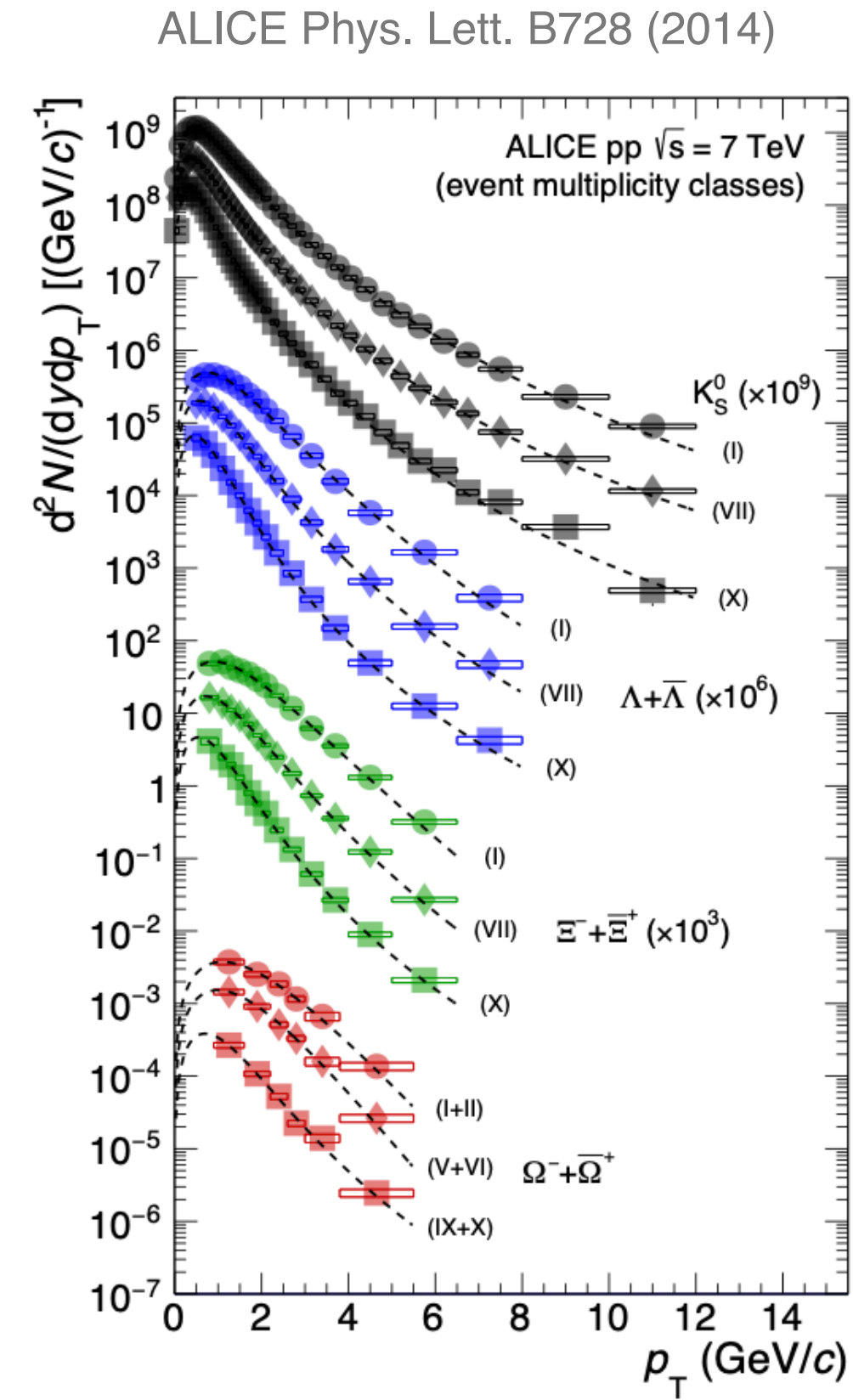


Fig. 1: p_T -differential yields of K_S^0 , $\Lambda + \bar{\Lambda}$, $\Xi^- + \bar{\Xi}^+$ and $\Omega^- + \bar{\Omega}^+$ measured in $|y| < 0.5$. The results are shown for a selection of event classes, indicated by roman numbers in brackets, with decreasing multiplicity. The error bars show the statistical uncertainty, whereas the empty boxes show the total systematic uncertainty. The data are scaled by different factors to improve the visibility. The dashed curves represent Tsallis-Lévy fits to each individual distribution to extract integrated yields. The indicated uncertainties all represent standard deviations.